

ESSENTIAL MATHEMATICS GRADE 10: STATISTICS 1

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.1: Statistics 1
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Ungrouped data – Frequency distribution tables for discrete data – Mean, mode and median of ungrouped data – Application of graphs of data – Representing data using bar graphs, line graphs and pie charts – Interpreting data from bar graphs, line graphs and pie charts – Representing data on a digital device – Project: collecting, organising, representing and interpreting data to guide school or community decisions
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) draw a frequency distribution table for ungrouped data b) determine mean, mode and median of ungrouped data c) represent data using bar graphs, line graphs and pie charts d) interpret data from bar graphs, line graphs and pie charts e) appreciate the importance of data organization and interpretation in making effective decisions
Core Competencies	– Critical thinking and problem solving: the learner develops interpretation and inference skills when collecting data, building frequency tables and drawing conclusions from graphs and charts. – Digital literacy: the learner develops the skills of connecting and interacting with digital technology when using digital devices to draw graphs and charts to represent the data collected.
Core Values	– Integrity: the learner displays honesty while working in groups and collecting data, such as when making frequency distribution tables. – Love: the learner is considerate of others when working with peers to extract information from graphs and charts and draw conclusions.
Science & Engineering Practices	– Asking questions and defining problems: learners ask what a school transport pattern or a favourite-colour poll can tell us and what data is needed to answer it. – Analyzing and interpreting data: learners organise data in frequency tables, compute mean, median and mode, and read graphs and charts to find patterns.

	– Using mathematics and computational thinking: learners calculate averages, convert frequencies to pie chart angles and use digital tools to plot graphs. – Developing and using models: learners build a cumulative data model (table, statistics and graphs) that is revised as new evidence arrives. – Engaging in argument from evidence: learners defend recommendations to the school using their graphs and statistics. – Obtaining, evaluating and communicating information: learners present findings and critique how others have represented data.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Decision Making): the learner makes informed decisions when identifying patterns from the graphs and charts and drawing conclusions.
Career Connections	– Journalism and media: reporting and presenting data-driven stories with graphs and charts. – Sports management and coaching: analysing player statistics such as race times and match scores of Kenyan athletes. – Hospitality industry: tracking customer preferences and sales of items such as mandazi, tea and nyama choma. – Sole proprietorship (kiosk or market stall trading): using sales data to decide what stock to buy. – Guidance and counselling: using survey data to understand learners' needs. – Photography and creative arts: using audience data to guide planning.
Focus for Lessons	This unit uses real data from the learners' own school, such as how they travel to school and which food or colour they prefer, to show why organising and interpreting data matters. Learners progress from frequency tables to mean, median and mode, then to bar graphs, line graphs and pie charts, finishing with a data project that guides a real decision in the school or community. Each lesson follows Predict, Observe, Explain, Driving Question Board and Model Building phases, with a class data model revised cumulatively.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision? KICD KEY INQUIRY QUESTIONS: 1. How do we use statistics in day-to-day life?
Anchoring Phenomenon	The school principal must decide how to use limited resources: whether to arrange a school bus route, build a bicycle shed, or add a second lunch serving. Different learners give conflicting opinions: 'Almost everyone walks', 'Most of us come by matatu', 'The bus is always full'. Students collect real data on how their class travels to school (walking, matatu, boda boda, bicycle, private car, school bus) and see a long, confusing list of 40 responses where nobody can say who is right. Once the same data is organised into a frequency table and shown as a bar graph and pie chart, the answer becomes clear, yet the mean, median and mode tell slightly different stories. The puzzle: how can the same raw list of numbers seem meaningless in one form and decisive in another, and which summary or graph should the principal trust?
Storyline Thread	Lesson 1: Anchoring phenomenon. Learners predict how the class travels to school, collect raw data, and see that the unorganised list does not settle the principal's question. The Driving Question Board is launched. Lesson 2: Frequency distribution tables. Learners use tally marks to build a frequency table for discrete data (travel mode, favourite colour, tree types in the compound) and compare it with the raw list. Lesson 3: Mode and mean. Learners predict the 'typical' value, then compute the mode and mean from the frequency table for ungrouped data such as the number of siblings or shoe sizes. Lesson 4: Median and comparing measures. Learners order the data, find the median, and discuss when mean, median or mode best represents the data (for example Kenyan athletes' race times).

	Lesson 5: Bar graphs and line graphs. Learners draw bar graphs for categories and line graphs for change over time (daily mandazi sales), and discuss which graph fits which kind of data. Lesson 6: Pie charts. Learners convert frequencies to angles, draw pie charts of the school travel data and family budgets, and compare representations for the principal's question. Lesson 7: Interpreting graphs and charts and using digital tools. Learners in pairs extract information, draw and defend conclusions, spot misleading graphs, and recreate graphs on a digital device. Lesson 8: Project and consolidation. Learners present data-based recommendations to the principal, revise the class model of how data becomes evidence, and return to the DQB to answer the driving question.
Supporting Phenomena	– Two Kenyan athletes have race times for five 800 m races; one has the faster mean but the other is more consistent, so which is the better pick for the Kericho county team? – A mandazi seller in Kisumu records daily sales for a week; a line graph shows a sharp rise on market day that the total alone hides. – A newspaper pie chart of how a Nairobi family spends its monthly income (rent, food, school fees, transport, savings) leads to different conclusions depending on how the slices are read. – The class tallies tree types in the school compound (grevillea, cypress, eucalyptus, jacaranda) and the frequency table reveals which species dominates before anyone counts them again.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (1 lesson); the DQB can be continued in a short follow-on slot if needed): Whose Claim Is Right? Launching the Principal's Problem

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon: Mr Otieno, the principal of Lakeview Senior School in Kisumu, has money for only one of three options (a school bus route, a bicycle shed or a second lunch serving), and learners give conflicting claims about how they travel to school. Learners predict the travel split, collect the 40 responses from Grade 10 Green and discover that an unorganised list cannot settle the argument. This creates the need to organise, summarise and display data.
Knowledge	By the end of the lesson the learner should be able to: (1) state that raw data is an unorganised list of responses; (2) explain why a claim such as 'almost everyone walks' needs evidence; (3) identify the six travel categories used by Grade 10 Green: Walk, Matatu, Boda boda, Bicycle, Private car, School bus; (4) explain why a long raw list is hard to use to decide between claims. Specific learning outcome link (KICD 3.1): collect data from the locality, such as means of transport used by learners to school; appreciate the importance of data organisation and interpretation in making effective decisions.
Skills	Making and recording a prediction with a reason (NGSS-style SEP: asking questions and defining problems; analysing and interpreting data). Collecting data accurately by recording one response per learner. Comparing a prediction with evidence. Asking testable questions. Communication and collaboration in pairs and groups. Core competencies: critical thinking and problem solving (interpretation and inference skills when collecting data); digital literacy is introduced later. PCI: Life Skills (Decision Making), shown when learners recognise that the principal needs evidence for an informed decision.
Attitudes	Integrity: learners record their own travel mode honestly and record others' responses accurately, even when it does not support their own claim. Love: learners are considerate of others' views when disagreeing about claims. Curiosity and openness to changing a prediction when evidence arrives.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 1 is the anchoring lesson. It builds only on prior Grade 9 knowledge of collecting data. Learners leave with an unsettled argument, a raw list of 40 responses, an open Driving Question Board and a first draft model. The totals are NOT revealed in this lesson. Lesson 2 will organise the same raw list into a tally and frequency table.
Safety Notes	No hazards. Handle the raw list sensitively: all names in the scenario are fictional, and learners may skip naming their own household transport details if uncomfortable (they only give one main travel mode). Avoid teasing about how anyone travels (for example private car versus walking). Remind learners that walking and cycling routes should be used safely and that no one should be ranked by how they travel. If learners have no devices, all tasks work on paper or the board.

B. LESSON OVERVIEW

Mr Kamau opens by telling the class that Mr Otieno, the principal of Lakeview Senior School in Kisumu, has money for only one option: a school bus route, a bicycle shed or a second lunch serving. Three learners disagree: Atieno says 'Almost everyone walks', Brian says 'Most of us come by matatu', and Zawadi says 'The bus is always full'. Learners first write a private prediction of how Grade 10 Green (40 learners) travels to school and compare predictions in pairs. They then collect the real data: each learner gives ONE main way of travelling (Walk, Matatu, Boda boda, Bicycle, Private car, School bus) and the 40 responses are recorded in learner order 1 to 40 on the board or on a shared sheet, producing a long, confusing raw list: 1 W, 2 M, 3 W, 4 S, 5 M, 6 W, 7 B, 8 W, 9 M, 10 C, and so on to learner 40. Learners try to settle the claims from the list alone and discover how hard this is: nobody can say quickly who is right, and the list does not give a clear picture. They record what they

notice and wonder, launch the Driving Question Board with the driving question 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?', sort questions into groups, and sketch a first draft model of how raw data might become evidence. No totals, percentages or graphs are shown; these are developed in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Categorical data example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Individuals, variables, and categorical & quantitative data Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners hear the principal's problem from Mr Kamau: Mr Otieno has money for only one of three choices (a school bus route, a bicycle shed or a second lunch serving). Learners read the three claims on the board: Atieno, 'Almost everyone walks'; Brian, 'Most of us come by matatu'; Zawadi, 'The bus is always full'. Individually, in their exercise books, they write: (a) which claim they think is closest to the truth for Grade 10 Green and why; (b) their own prediction of how many of the 40 learners use each mode: Walk, Matatu, Boda boda, Bicycle, Private car, School bus (the six numbers should add to 40); (c) which of the three options they think the principal should choose, and what evidence they would want. They then compare predictions with a partner and notice that predictions differ. Learners are not told the real totals.	Read the scenario aloud slowly, then say: 'Mr Otieno cannot buy everything. He can fund ONE thing. What would you want to know before you advise him?' WAIT TIME 10-15 seconds before taking any answer. Display the three claims on the board without saying which is correct. Say: 'Write your prediction silently for 3 minutes. There is no wrong prediction today; it is your starting idea.' Circulate and look for learners who write numbers that do not add to 40; prompt with 'How many learners are there in total in our class?' Then pair learners for 2 minutes and cold-call 4 pairs (not volunteers) to share ONE number they disagreed on. Record 3 to 4 predicted Walk numbers on the board (for example 10, 20, 25, 30) without comment, and ask: 'Why do we not agree if we are all in the same class?' WAIT TIME 10 seconds. Do not reveal or hint at the actual counts.	Strategy: Prediction with reasoning (commit-before-evidence) plus Think-Pair-Share. Writing a private prediction first makes every learner hold a stake in the question, so the later raw list matters to them. Seeing different predictions side by side on the board shows that opinions alone cannot settle the argument, which creates the need for evidence. This links directly to the anchoring phenomenon: different learners, conflicting claims, one principal.	Look for: (1) each learner writes a prediction with a reason, not only a guess; (2) predictions for the six modes add to 40, or the learner corrects them after prompting; (3) at least 3 pairs name a type of evidence they would need, such as 'count everyone' or 'ask each learner'; (4) learners use the six mode names correctly. Quick check by show of hands: 'Who changed their mind after hearing a partner?' Note learners who rely only on feelings ('I just think') for follow-up in the later phases.
Observe Phase  VIDEO: Identifying individuals, variables and categorical variables in a data set Source: Khan Academy (English	Learners collect the real data. Each learner answers ONE question: 'What is your main way of travelling to school?' choosing from Walk, Matatu, Boda boda, Bicycle, Private car, School bus. Responses are recorded in learner order 1 to 40 on a shared class sheet or the board, in the order of the register. The raw list reads: 1	Say: 'We are now collecting evidence the way a real researcher would. Each person gives ONE main way of travelling, the way you travel MOST days.' Stress integrity: 'Record what is true, not what makes your claim win.' Call learners 1 to 40 in register order and write each response in the order given, using	Strategy: Evidence collection with a 'noticing and wondering' routine. Learners collect real data about themselves, so the evidence feels authentic. Working from the raw list forces them to meet the problem directly: the data exists but is not usable. Naming the difficulty ('it is long', 'I lost count', 'no order') builds the need for	Look for: (1) learners record their own response honestly and accurately; (2) the class list has exactly 40 entries with codes matching the six modes; (3) learners can describe the difficulty in their own words (for example 'too long to count by eye'); (4) at least 4 learners state that they cannot be sure which claim is

- US curriculum) Search ARES for similar videos EXERCISE: Individuals, variables, and categorical & quantitative data Source: Khan Academy (English - US curriculum) Search ARES for similar readings	W, 2 M, 3 W, 4 S, 5 M, 6 W, 7 B, 8 W, 9 M, 10 C, 11 W, 12 M, 13 Bi, 14 W, 15 M, 16 W, 17 S, 18 M, 19 W, 20 B, 21 M, 22 W, 23 C, 24 W, 25 M, 26 S, 27 W, 28 Bi, 29 M, 30 W, 31 B, 32 M, 33 W, 34 C, 35 S, 36 M, 37 W, 38 Bi, 39 B, 40 M. In pairs, learners copy the list and try to answer: (a) Is Atieno right that almost everyone walks? (b) Is Brian right that most of us come by matatu? (c) Does the list tell us whether the school needs a bus route, a bicycle shed or a second lunch serving? They record what they notice and what makes it hard (for example, they have to scan 40 items, they lose count, answers are in no order).	the codes W, M, B, Bi, C, S. Have a learner scribe check each entry aloud. Once the list is complete, say: 'You have 4 minutes with your partner. Use only this list. Decide whether Atieno or Brian is right, and tell me how sure you are.' Do not teach tallying or tables; if a learner starts tallying, say: 'Interesting, hold that thought; first tell me how easy or hard it is.' Circulate and note who tallies, who loses count, and who gives up. WAIT TIME 10-15 seconds after asking 'How sure are you?' Cold-call 5 learners to give a verdict and one reason. Then ask: 'Can you tell Mr Otieno, in one sentence, who is right?' Let the silence stand. Do NOT reveal the totals or the answer.	organisation and shows that data alone is not evidence until it is organised. This links to the phenomenon: the same list that seems meaningless can become decisive once organised.	right; (5) learners who begin to group or tally by themselves are noted for Lesson 2 but not stopped. Check notebooks for a copied list with no mistakes in order.
Explain Phase VIDEO: Categorical data example Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Trends in categorical data <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners discuss in groups of 4 what they can and cannot say from the raw list. They return to their predictions and the three claims. They decide for each claim whether the list lets them judge it: Atieno ('Almost everyone walks'), Brian ('Most of us come by matatu'), Zawadi ('The bus is always full'). They discuss whether the class data can speak about Zawadi's claim at all (the school has a single 14-seater bus serving the whole school, but this class list is only Grade 10 Green, so it gives limited information about that claim). They write a short statement: 'The raw list helps us by ____, but it does not help us because ____.' They also say what the principal would want before choosing one of the three options, and why one class may not be enough to speak for the whole school.	Say: 'Turn to your group. Take 4 minutes. For each claim, can the list prove it, disprove it, or can we not tell? Give a reason.' Circulate and ask: 'What would you need to do to the list to be sure?' Press for reasoning: 'You said it looks like many walk. What does "many" mean to Mr Otieno?' Probe Zawadi's claim: 'Zawadi says the bus is always full. Which part of our list could confirm that?' WAIT TIME 10-15 seconds, then cold-call 3 groups. Introduce the key ideas in plain words only: 'raw data is the unorganised list; a claim needs evidence; one class is a small group compared with the whole school.' Do not give totals, frequency tables, percentages, or mean, median or mode (these belong to later lessons). Close with: 'So we have data but no decision. What is missing?' Collect 4 to 5 responses on the board under the heading 'What we need'.	Strategy: Claim-Evidence-Reasoning (CER) discussion in groups. Learners judge each claim against the data, state what the evidence can and cannot support, and give a reason. This builds the habit of separating opinion from evidence, which is the core idea behind the driving question. Distinguishing what the class data can say from what it cannot (for example the bus claim, or the whole school) develops careful reasoning about sample size without using technical terms. It connects to the phenomenon because conflicting claims remain unresolved by the raw list.	Look for: (1) groups give reasons using the list rather than feelings; (2) at least 3 groups say 'we cannot tell' for at least one claim and explain why; (3) learners use the term 'evidence' correctly; (4) at least 2 groups notice that one class is not the whole school; (5) learners can state what is missing (a clearer organised summary). Use a quick thumbs-up/down check: 'Can the raw list alone settle Atieno and Brian's argument?' Expect most thumbs down. Follow up with learners whose thumbs are up, asking them to show how they know.

Driving Question Board (DQB) Creation  VIDEO: Identifying individuals, variables and categorical variables in a data set Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Individuals, variables, and categorical & quantitative data Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners help launch the Driving Question Board. The driving question is posted at the top: 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' Each learner writes 2 questions on sticky notes or paper strips from what they have seen (for example 'How many of us walk?', 'How can we count quickly without mistakes?', 'Is one class enough?', 'Is it fair to decide on the bus from this?', 'What is the best way to show the results?'). In groups, they cluster similar questions and agree on 3 to 4 category headings. The class posts a key inquiry link: 'How do we use statistics in day-to-day life?' Learners also add the three claims (Atieno, Brian, Zawadi) and the raw list to the board as the starting evidence.	Post the driving question and read it aloud. Say: 'This is our big question for the next lessons. Write 2 questions of your own that you want answered. They can be small or big.' Give 3 minutes of silent writing. Then say: 'Bring your notes up and stick them where they belong. If a question looks like another one, put them together.' Cold-call 5 learners to read one question aloud and say why it matters. Help the class form clusters such as 'Counting and organising', 'Summaries and averages', 'Graphs and pictures', 'Fairness and trust' and 'Whole school'. WAIT TIME 10-15 seconds before asking: 'Which questions do you think we can answer first?' Mark with a star the questions about counting and organising. Say: 'We will return to this board at the end of every lesson.' Do not answer the questions now; note the ones on mean, median and mode to revisit later.	Strategy: Driving Question Board with question clustering. Learners turn their confusion into questions they own, and clustering shows patterns in what the class wants to know. The board becomes a visible record of understanding that grows from lesson to lesson, so learners can see their questions being answered. It connects to the phenomenon because every question traces back to the unresolved problem of how to turn the messy list into a fair decision for Mr Otieno.	Look for: (1) each learner posts at least 2 questions; (2) at least half the questions are about organising, summarising or showing data rather than yes/no or off-topic questions; (3) learners can justify the cluster headings; (4) at least 3 learners ask about fairness or whether the data is enough (for example one class versus the whole school); (5) learners connect questions to the principal's decision. Note learners who write only 'what is statistics?' for gentle follow-up. Photograph or keep the board for use in later lessons.
Model Building Phase  VIDEO: Mosaic plots and segmented bar charts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Individuals, variables, and categorical &	Learners draw a first draft model in their books to show their current thinking about how a messy list of data could become evidence for the principal. The model includes: (1) the messy raw list of 40 responses; (2) a box or arrow labelled with their idea of what should happen next (for example 'count', 'sort', 'draw something'); (3) the output they expect (a clear answer for Mr Otieno); (4) a question mark where they are unsure. They add a short caption: 'What I think will make the data useful is ____.' Pairs swap models and give one comment each. Learners also complete the first column of a summary table: 'What	Say: 'Draw how you think a messy list could turn into clear evidence for Mr Otieno. Use boxes and arrows. It is fine if you are unsure; put a question mark where you do not know.' Model with a quick sketch of two boxes and one arrow without filling in the steps. Circulate and ask: 'What happens to the 40 responses in your model?' and 'How will you know your answer is right?' WAIT TIME 10-15 seconds after each question. Ask pairs to swap and say one thing they like and one question they have. Cold-call 3 learners to share their 'what I think will make it useful' caption. Do not correct the models; say: 'We will	Strategy: Initial model drawing (cumulative model building). Drawing a first draft makes learners' ideas visible, including gaps and misconceptions, and gives a baseline to compare with later. Because the model will be revised after each lesson, learners see how their understanding develops from raw list to organised table, summaries and graphs. It links to the phenomenon as the model explains how a messy list could become a decisive piece of evidence.	Look for: (1) the model shows the raw list leading to some organised or summarised output; (2) learners include at least one action (count, sort, group, draw); (3) learners mark uncertainty with a question mark or note; (4) the caption refers to evidence for the principal; (5) peer comments are specific. Use a quick exit prompt: 'In one sentence, why can we not decide from the raw list alone?' Check that at least most learners write about the list being long, unorganised or hard to count.

quantitative data Source: Khan Academy (English - US curriculum) Search ARES for similar readings	we observed' (a 40-response list, conflicting claims, no clear answer).	revise these models at the end of the next lessons.' Collect the models or tell learners to keep them safely in their books. Remind learners that tomorrow they will start organising the same list.		
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D. TEACHER REFLECTION

After the lesson, consider: Did learners commit to a prediction before seeing the data, and did the differing predictions create real curiosity? Did the raw list feel genuinely confusing, so learners saw the need for organisation, or did some learners quickly tally it and settle the argument? If so, how will you build on this in Lesson 2? Which learners relied on opinion rather than evidence? Were all learners comfortable giving their travel mode honestly, with no teasing? Did learners notice the limits of one class and the point that the bus claim cannot be judged from this class list? Which questions on the DQB are most useful for Lesson 2 (tally and frequency tables)? Keep the board, the raw list and the draft models for revisiting in later lessons. Did you avoid revealing the totals?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did we observe? We collected 40 responses from Grade 10 Green about main travel mode (Walk, Matatu, Boda boda, Bicycle, Private car, School bus), saw three conflicting claims from Atieno, Brian and Zawadi, and found the long raw list was hard to use to say who is right.
What did I learn?	What did we learn? Raw data is an unorganised list; a claim needs evidence; a long list does not settle an argument by itself, and one class is not the whole school.
How does this explain the phenomenon?	How can we explain it so far? We have data but not yet evidence for Mr Otieno because the list is unorganised. We predict that organising and showing the data will make the answer clearer, but we still need to find out how, and which summary or graph to trust.

LESSON 2 (40 minutes (one lesson; the 5 phases are done in sequence, with a short extension for the DQB and model)): Tally and Frequency Tables: Making Sense of the 40 Responses

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners organise the raw list of 40 travel responses into a tally and frequency table, check that the frequencies add to the total of 40, and judge how far the table helps settle the claims about how the class travels to school.
Knowledge	By the end of the lesson the learner should be able to: (a) define raw data, tally marks and frequency; (b) draw a frequency distribution table (Category, Tally, Frequency) for ungrouped categorical and discrete data; (c) check that the frequencies add up to the total number of responses; (d) calculate the percentage for each category using $f / n \times 100$.
Skills	Collecting and organising data, tallying in groups of five, checking totals, reading a table to extract information, comparing two representations of the same data (raw list and table), and communicating a conclusion using evidence. NGSS Science and Engineering Practices embedded: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations.
Attitudes	Integrity: honesty while working in groups and making frequency tables (no adjusting counts to fit a favourite claim). Love: consideration of others when working with peers to extract information and draw conclusions. Appreciate the importance of data organisation and interpretation in making effective decisions. Core competencies: Critical thinking and problem solving (interpretation and inference skills when collecting data). PCI: Life Skills (Decision Making).
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 1 left learners with a raw list of 40 responses from Grade 10 Green and three claims (Atieno: 'Almost everyone walks'; Brian: 'Most of us come by matatu'; Zawadi: 'The bus is always full') that the unorganised list could not settle. In this lesson learners build the frequency table (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4 = 40) and see for the first time that the same data becomes readable once organised. This gives the evidence base that Lessons 3 to 7 will summarise (mean, median, mode) and display (graphs and pie charts) before the Lesson 8 recommendation to Mr Otieno. The lesson does not yet teach mode, mean, median or graphs.
Safety Notes	Low-risk classroom activity. Learners share the class list respectfully and never tease a classmate about how they travel (for example boda boda or walking long distances). Names are not attached to answers on the display; use learner numbers 1 to 40 only. Pencils and rulers are handled carefully, and learners keep walkways clear when moving to the board.

B. LESSON OVERVIEW

Mr Kamau returns the raw list of 40 travel responses from Lesson 1 (learner numbers 1 to 40, coded W, M, B, Bi, C, S). Learners first predict what a more organised version might show, then work in groups of four to tally the 40 responses into five-bar-gate tally marks, count frequencies and check that they add to 40. They compare the raw list with the table and test the claims of Atieno, Brian and Zawadi against the evidence. Walk 14 and Matatu 12 are the largest groups, neither is more than half, and together they make 26 of 40, which is 65%. Learners then apply the same skill to a new data set (30 trees in the Lakeview compound: Grevillea 12, Cypress 8, Eucalyptus 6, Neem 4) and discuss discrete data. They add new questions to the Driving Question Board and revise their first draft model: raw data must be organised before it can be used as evidence. Zawadi's claim about the bus cannot be tested with this class table because the school has one 14-seater bus serving the whole school, which is a question for the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading dot plots & frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the raw list from Lesson 1 displayed on the board (1 W, 2 M, 3 W, 4 S ... 40 M). Individually, then in pairs, they write a prediction in their books: 'If we count how many learners use each way of travelling, which way will have the most learners, and about how many? Will anyone be right: Atieno, Brian or Zawadi?' They also predict: 'What would we have to do to the list to see the answer quickly?' Pairs share their predicted counts for Walk and Matatu, and write them in a prediction column to be checked later.	Display the raw list in learner order. Say: 'Yesterday we could not settle Atieno's and Brian's argument from this list. Before we do any counting, write down your best prediction: which mode of travel will have the most learners, and how many?' Give 1 minute silent writing, then WAIT TIME of 10 seconds after asking: 'What could we do to this list to make it easier to read?' Cold-call 3 learners for their predicted counts and 2 learners for ideas on organising (listen for 'count', 'group', 'table'). Record predictions on the board in a Prediction column without judging. Say: 'We will check these predictions with evidence before the lesson ends.'	Prediction with commitment (Predict-Observe-Explain): learners put a numerical prediction on record, so that comparing it with the actual frequency table creates cognitive conflict or confirmation. Naming the need to organise the list helps learners see why a table is needed, and links back to Lesson 1's unsettled claims.	Observe whether learners write a specific numerical prediction rather than just a category. Look for: (1) predictions that name a category and an approximate count; (2) learners who connect to a claim (Atieno, Brian or Zawadi); (3) ideas about counting or grouping. Note the 3 or 4 learners who predict that nobody can know without counting; they have the right idea for the Observe phase.
Observe Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Tables to Organize and Display Data (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of four, learners receive a copy of the raw list (learner numbers 1 to 40 with codes W, M, B, Bi, C, S). Step 1: draw a table with three columns: Category, Tally, Frequency, and six rows in the order Walk, Matatu, Boda boda, Bicycle, Private car, School bus. Step 2: one learner reads the list aloud, response by response, while another puts a tally mark in the correct row, grouping every five marks (four strokes crossed by a fifth). Step 3: each group counts tallies to fill the Frequency column, then adds the six frequencies to see whether they equal 40. Step 4: groups whose total is not 40 find the mistake (a missed or double-counted response) by re-reading the list. The final agreed table is Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School	Model the first 5 responses on the board: 'Learner 1 is W, so I put one stroke in the Walk row. Learner 2 is M, one stroke in Matatu.' Show how the fifth stroke crosses the four. Circulate and ask: 'How do you know you have not missed learner 23?' and 'What should the six frequencies add up to? Why?' Insist on one reader and one tally keeper, switching roles halfway so every learner does both. If a group's total is not 40, do not give the error; say 'Your total is 39. Where could a response have gone missing? Re-read together.' Give groups a WAIT TIME of 10 to 15 seconds before offering a hint. After 10 minutes, ask 4 groups (cold-call by group number) to read their frequencies; write them on the board and resolve any disagreement by tallying the	Data organisation as evidence-building: learners physically transform an unorganised list into a table and use the total check (frequencies must add to 40) as a built-in error detector. The side-by-side comparison of the raw list with the table builds understanding that organisation, not new data, creates the clarity.	Observable indicators: (1) tally marks are grouped in fives correctly; (2) the six rows appear in the order Walk, Matatu, Boda boda, Bicycle, Private car, School bus; (3) frequencies are 14, 12, 4, 3, 3, 4; (4) the group checks that they sum to 40 without being prompted; (5) percentages are computed as $f / 40 \times 100$. Use a quick checklist per group. Common errors to watch for: counting the learner number instead of the response; ending with a total of 39 or 41; tallying with five single strokes instead of a gate; forgetting that Bicycle (Bi) and Boda boda (B) are different rows.

	bus 4 = 40. Step 5: groups add a Percentage column using $f / n \times 100$: Walk 35%, Matatu 30%, Boda boda 10%, Bicycle 7.5%, Private car 7.5%, School bus 10% (sum 100%). Learners compare the table to the raw list and record two things the table shows that the list did not.	disputed rows together. Remind learners to be honest and report what the list really says, even if it contradicts their own prediction (Integrity). Remind them to use learner numbers, not names.		
Explain Phase  VIDEO: Two-way relative frequency tables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Frequency Tables to Organize and Display Data Practice Source: CK-12  Search ARES for similar readings	Learners use the completed table to test the three claims. In pairs they answer: (a) Is Atieno right that 'almost everyone walks'? Use the number and percentage. (b) Is Brian right that 'most of us come by matatu'? (c) Can the table settle Zawadi's claim that 'the bus is always full'? Learners find that 14 of 40 (35%) walk, which is the largest group but not almost everyone; 12 of 40 (30%) use a matatu, so not most; walk and matatu together are 26 of 40, 65%. For the bus, only 4 of 40 learners in Grade 10 Green use it, but Zawadi's claim is about the school's single 14-seater bus serving the whole school, so this class table cannot confirm or refute it. Learners then practise on new data: the 30 trees in the Lakeview compound (raw list given: for example G, C, G, E, N ...) and complete a tally and frequency table to get Grevillea 12, Cypress 8, Eucalyptus 6, Neem 4 = 30. They write one sentence in the form 'The most common tree is ___ with ___ out of ___.' and decide how the tree data differs from travel data. Discussion on discrete data: counts such as number of siblings are numbers, while travel mode and tree type are categories; both can go into the same Category/Value, Tally, Frequency layout. Finally, learners	Ask: 'Using only the table, is Atieno right that almost everyone walks? Give me a number.' Give WAIT TIME of 10 to 15 seconds, then cold-call 3 pairs for each claim. Press for evidence: 'What does 14 out of 40 tell you? Is that more than half? How do you know?' For Zawadi: 'Our table says 4 learners use the bus. Does that prove the bus is not full?' Wait 10 seconds. Guide to: the bus is a single 14-seater serving the whole school, so one class cannot confirm or refute it; we would need data from the whole school. Do not tell learners the answer to the whole-school question. For the tree practice, supply the raw list of 30 responses and let pairs work independently for 6 minutes; cold-call 2 pairs to read their frequencies, and have the class check that the total is 30. Ask: 'How is travel mode different from number of siblings? Which one is a count?' Introduce the terms categorical and discrete data clearly and briefly, and say 'We are not yet finding the mean, median or mode; that comes next lesson.' Praise precise evidence-based sentences: 'Good, you used the number 14 out of 40 to support your claim.'	Claim-Evidence-Reasoning (CER): learners state a claim about each learner's statement, cite specific frequencies and percentages as evidence, and give reasoning (for example 'more than half would be more than 20 learners'). Transferring the same skill to the tree data checks understanding of the table-building procedure in a new context, and the categorical versus discrete contrast develops understanding of data types.	Listen for CER statements that cite numbers (14 of 40, 12 of 40, 26 of 40, 65%) and use 'more than half' reasoning (more than 20). Check that learners do not claim the table proves anything about the school bus being full. Collect the tree tables: correct frequencies are 12, 8, 6, 4 summing to 30. Exit-ticket style check on whiteboards: 'Write the frequency and percentage for Matatu.' Expected: 12 and 30%. Note learners who still confuse frequency with percentage for re-teaching in small groups.

	revisit their predictions and mark whether each was close.			
Driving Question Board (DQB) Creation  VIDEO: Analyzing trends in categorical data <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Creating frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1, which displays: 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' Each learner writes one new question or one new answer on a sticky note or paper slip. Examples: 'Is one class enough to decide for the whole school?' 'How can we show the table as a picture?' 'Which number best represents how we travel?' 'How full is the school bus if only 4 of us use it?' Learners place notes on the board and, as a class, sort them into groups: ones answered by the table (for example how many walk), ones to be answered in later lessons (summaries and graphs), and ones that need more data (the whole school and the bus). Learners mark a note answered: 'Which mode of travel is most common in our class? Walk, 14.'	Say: 'Put one thing we can now answer and one new question on the board. Use learner numbers, not names, if you mention classmates.' Give 1 minute of silent thinking and WAIT TIME of 10 seconds before they write. Cold-call 4 learners to read and place their notes and say why. Ask: 'Which of these questions can the table already answer? Which need something more than a table?' Move the answered notes to an 'Answered by evidence' column and label two columns: 'Need a typical value or graph' and 'Need whole-school data'. Point out that Zawadi's bus claim sits in the last column. Preview: 'Next lesson we will find one typical value for our data.' Do not answer future questions now.	Question sorting on the DQB: learners categorise questions by the type of evidence needed. This tracks growing understanding and builds a visible link between what has been learned (frequency table), what is still unknown (typical value, graphs, whole school) and the final recommendation for Mr Otieno.	Check whether each learner posts a relevant question or answered statement and whether they place it in a sensible group. Look for questions that show awareness that one class is a small sample, or that the table needs a picture. Note learners whose questions are only 'what is a tally?' and who need support. Record 3 strong questions to guide planning for Lesson 3.
Model Building Phase  VIDEO: Mosaic plots and segmented bar charts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading dot plots	Learners revisit the first draft model from Lesson 1 (a diagram or flow chart showing how data goes from collection to a decision for the principal). Individually or in pairs, they add a new step between 'collect raw data' and 'decide': 'Organise into a tally and frequency table, then check the frequencies add up to the total.' They add an example box: 'Travel: Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4 = 40.' They also write a limits note: 'One class (Grade 10 Green) is a small sample; the table alone does not choose between a	Say: 'Open your model from Lesson 1. What step is missing between collecting the 40 responses and advising Mr Otieno? Add it now with the numbers.' Give 2 minutes of silent work, then WAIT TIME of 10 seconds before asking: 'What did you add and why?' Cold-call 3 learners to explain their new step. Ask a probing question: 'Does the table tell Mr Otieno what to build?' Guide learners to the limits: it shows which groups are bigger, but they still need typical values and graphs and data from the whole school. Collect a photo or	Cumulative model revision: learners add a new mechanism (organising data into a frequency table with a total check) to their evolving model of how raw data becomes evidence. Writing a limits note supports scientific reasoning about uncertainty and sample size.	Look for: (1) the new step is placed between collecting and deciding; (2) it mentions tally, frequency and the total check; (3) the example uses correct travel numbers (14, 12, 4, 3, 3, 4 = 40); (4) a limits note on sample size. Use a 3-level quick check: step present but no numbers; step with numbers; step with numbers and limits note. Plan a short re-teach for learners at the first level.

& frequency tables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	bus, a shed or a second lunch serving.' Learners compare their updated models with a neighbour and fix any missing steps. Optionally they complete the sentence: 'The same list looks meaningless as raw data but clear as a table because ____.'	note of 3 models to plan the next lesson. Close with: 'Next lesson we will find a typical value for our data.'		
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did every group get a total of 40 on its own, and which strategies did they use to find a missing or double-counted response? (2) Which learners still mix up frequency and percentage, or tally with five single strokes instead of the gate? (3) Did learners avoid claiming that the table proves anything about the school bus, and recognise that one class is a small sample? (4) Did the comparison between the raw list and the table help learners see why organisation matters? (5) Were the learners' predictions close to 14 and 12, and how did they react to the evidence? Note which DQB questions to carry into Lesson 3 (typical value: mode and mean) and which learners need support with the Category, Tally, Frequency layout. Ensure learner names stay out of the displayed data.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tallied the 40 raw responses from Grade 10 Green and got Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4, which add to 40. Percentages are Walk 35%, Matatu 30%, Boda boda 10%, Bicycle 7.5%, Private car 7.5%, School bus 10%. For the 30 trees in the Lakeview compound we got Grevillea 12, Cypress 8, Eucalyptus 6, Neem 4.
What did I learn?	A frequency table has the columns Category (or Value), Tally and Frequency. Tally marks are grouped in fives, and the frequencies must add up to the total number of responses. Percentage = $f / n \times 100$. Categorical data (travel mode, tree type) and discrete data (counts such as siblings) can both be organised this way.
How does this explain the phenomenon?	The raw list could not settle the argument, but the table shows Walk (14 of 40, 35%) is the largest group and Matatu (12 of 40, 30%) is second. Neither is more than half, so 'almost everyone walks' and 'most come by matatu' are both wrong, although together they make 26 of 40, which is 65%. Only 4 of 40 use the school bus and 3 of 40 cycle. The table cannot settle Zawadi's claim about the single 14-seater school bus, and one class is a small sample, so we need typical values, graphs and whole-school data before advising Mr Otieno.

LESSON 3 (40 minutes (1 lesson)): Typical Value: Mode and Mean

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find the mode from a frequency table and calculate the mean using $\Sigma fx \div \Sigma f$ for ungrouped data, so they can describe a 'typical' value in Grade 10 Green's data for Mr Otieno.
Knowledge	By the end of the lesson, the learner should be able to: (a) determine the mode of ungrouped data from a frequency table (the value or category with the highest frequency); (b) calculate the mean of ungrouped data using $\Sigma fx \div \Sigma f$ from a frequency table with an fx column; (c) explain why the mode can be used for categories (travel mode = Walk, 14) while the mean applies only to numbers (siblings: mean $83 \div 40 = 2.075$, about 2.1 siblings; mode 2).
Skills	Reading and extending a frequency table (adding an fx column); multiplying and adding systematically; checking that Σf equals the total of 40; rounding to 1 decimal place and giving units (siblings); constructing a written explanation of what 'typical' means. NGSS-aligned practices: Analysing and interpreting data; Using mathematics and computational thinking; Constructing explanations.
Attitudes	Integrity: the learner records and calculates honestly and checks results with peers. Love: the learner is considerate when partners make errors and helps them correct these. Appreciation of how summarising data helps in making effective decisions. Core competency: Critical thinking and problem solving (interpretation and inference). PCI: Life Skills (Decision Making).
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 1 showed that the raw list of 40 travel responses could not settle the claims. Lesson 2 organised the data into a frequency table (Walk 14; Matatu 12; Boda boda 4; Bicycle 3; Private car 3; School bus 4). Lesson 3 asks 'what is typical for our class?' Learners find the mode of travel (Walk) and the mode and mean of a numerical set (siblings of the same 40 learners). They see that one number can summarise a table, but that the mean fits only numbers. Lesson 4 will add the median and ask which measure is best.
Safety Notes	No hazards. Calculators are optional and should be shared fairly. Siblings data is sensitive: families differ, so learners give only a number, no names or comments about family size, and respect for classmates is stressed.

B. LESSON OVERVIEW

Mr Otieno asks, 'What is the typical Grade 10 Green learner at Lakeview Senior School in Kisumu?' Learners first predict a typical travel mode and a typical number of siblings. They then use the frequency table from Lesson 2 to find the mode of travel (Walk, 14 of 40). Next they work with the siblings table (0 siblings 5; 1 sibling 9; 2 siblings 12; 3 siblings 8; 4 siblings 4; 5 siblings 2; total 40), find the mode (2) and build an fx column to calculate the mean ($0+9+24+24+16+10 = 83$; $83 \div 40 = 2.075$, about 2.1 siblings). They discuss why a mean of 2.1 siblings is not a real family, why the mean cannot be found for travel mode, and what these summaries do and do not tell the principal. The median and the choice of best measure are left for Lesson 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(About 7 minutes) Learners are reminded of the principal's problem	Write on the board: 'Mr Otieno asks: what is a TYPICAL Grade 10	Prediction with a public record (Think-Pair-Share). Writing a	Observe: Do learners give reasons, not just guesses? Do at

 VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	and of Lesson 2's frequency table. Individually, on scrap paper, they write: (1) the 'typical' way a Grade 10 Green learner travels to school; (2) their prediction of the 'typical' number of siblings a learner in the class has, and why. They are told the class's siblings counts were recorded as a frequency table. They pair up and compare, then 3-4 pairs share predictions. Learners can see that some predicted a whole number and some a decimal.	Green learner?' Say: 'Without calculating, write down the typical way we travel to school and the typical number of siblings. You have 2 minutes of silent thinking.' WAIT TIME: 10-15 seconds after asking, 'What does the word typical mean to you?' before taking answers. Cold-call 4 learners (not volunteers) for their sibling prediction and record all predictions as tally marks on the board. Ask: 'Who thinks typical means the most common? Who thinks it means the middle or fair share?' Do not correct yet. Say: 'Keep your prediction. We will test it with the data.'	prediction first, then comparing, makes learners commit to an idea and notice that 'typical' can mean different things, which creates the need for precise measures (mode and mean).	least 3 different meanings of 'typical' (most common, middle, share) come out? Note learners who predict travel mode 'Walk' or 'Matatu' using Lesson 2's table. Look for the prediction that 'typical' is a fair share, a hint of the mean. Record the class range of sibling predictions on the board for later comparison.
Observe Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the mean: data displays Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(About 13 minutes) Part A (4 min): From the Lesson 2 travel table, learners underline the highest frequency and find the mode: Walk (14). They check that the frequencies add to 40. Part B (9 min): In pairs, they receive the siblings frequency table (0 siblings 5; 1 sibling 9; 2 siblings 12; 3 siblings 8; 4 siblings 4; 5 siblings 2). They (1) check $\Sigma f = 40$, (2) find the mode (2 siblings, $f = 12$), (3) add a third column fx (value $\times$ frequency): 0, 9, 24, 24, 16, 10, (4) add the fx column to get $\Sigma fx = 83$, (5) divide $83 \div 40 = 2.075$, about 2.1 siblings. They compare the result with their prediction.	Part A: 'Look at last lesson's travel table. Which group has the highest frequency? Show me with your finger.' WAIT TIME 10 seconds. Cold-call 2 learners: 'What is the mode of travel? How do you know?' Write: 'Mode = value or category with the highest frequency.' Part B: Say: 'The mode tells us which is most common. To share all the siblings out fairly, we need the mean. Add a new column called fx.' Model ONE row on the board: '3 siblings $\times$ 8 learners = 24 siblings. Why multiply?' WAIT TIME 10-15 seconds. Circulate with the prompt: 'Where is your check that the f column adds to 40?' Watch for learners who add the x values ($0+1+2+3+4+5 = 15$) or divide by 6 rows instead of 40; ask, 'How many learners are there altogether? So what do we divide by?' Ensure answers are rounded to 1 d.p. with units. Write: 'Mean = $\Sigma fx \div \Sigma f$'. Use the word 'mean', never 'average'.	Structured table construction (extending the frequency table with an fx column) and Notice and Wonder. The fx column makes the repeated-addition idea visible, so learners see why the mean needs frequencies, linking to Lesson 2's f column. Contrasting the mode (a count) with the mean (a share) shows two meanings of 'typical'.	Check observable indicators: (1) $\Sigma f = 40$ is checked before calculating; (2) fx values are 0, 9, 24, 24, 16, 10; (3) $\Sigma fx = 83$; (4) the divisor is 40, not 6; (5) the mean is written as 2.1 siblings; (6) the mode of siblings is given as 2, not 12. Tally how many pairs reach 83. Pairs who get a different Σfx must find their own error by checking each row. Note common errors: giving the mode as the frequency (12), or ignoring the 0 siblings row.
Explain Phase	(About 10 minutes) Learners work	Ask: 'Nobody has 2.1 siblings. Is	Claim-Evidence-Reasoning (CER)	Look for explanations that (a) say

 VIDEO: Impact on median & mean: increasing an outlier <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Ungrouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	in groups of 4 to answer: (1) The mean is 2.1 siblings, but no learner has 2.1 siblings. What does 2.1 mean? (2) The mode of travel is Walk; why can we not calculate a mean of travel mode? (3) Mode 2 and mean 2.1 for siblings are close; Walk is the mode of travel, but it is only 14 of 40. Does 'Almost everyone walks' hold? (4) Mini-problem: Mama Achieng's mandazi sold on 5 days are 40, 45, 42, 50 and 45. Find the mean ($222 \div 5 = 44.4$ mandazi) and the mode (45). Each group writes a claim-evidence-reasoning statement for Mr Otieno: 'The typical Grade 10 Green learner ... because ...'.	the mean wrong? Talk in your groups.' WAIT TIME 15 seconds before sharing. Cold-call 3 groups (not volunteers). Use the revoicing prompt: 'Who can say that again in different words?' Guide: 'Imagine all 83 siblings shared out equally among 40 learners: how many each?' Push the link to the phenomenon: 'Walk is the mode with 14. Is 14 out of 40 almost everyone? What fraction, as a percentage?' (35%, from the Lesson 2 table.) Say: 'Walk is the most common group, but 35% is not more than half. The mode names the biggest group; it does not say it is the majority.' Provide the sentence frame: 'The typical learner ... The evidence is ... This helps Mr Otieno because ...'. Note that the median, and which measure is best, come in Lesson 4: 'Hold that question for next lesson.'	with group discussion. Learners must justify a claim using the mode and mean as evidence and reason about what each measure can and cannot show. Contrasting categorical and numerical data builds understanding of when each measure applies.	the mean is a fair-share value that need not be a whole number; (b) say the mean needs numbers, so it cannot apply to Walk, Matatu and so on; (c) cite 14 of 40 (35%) to show that the mode is not a majority; (d) get 44.4 mandazi and 45 correctly. Use a 3-level check: Meets = correct calculations and one correct explanation; Exceeds = also explains the limits of both measures with data from the lesson; Approaching = calculations only. Note groups for follow-up.
Driving Question Board (DQB) Creation  VIDEO: Means and medians of different distributions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Ungrouped Data to Find the Mean Practice Source: CK-12	(About 5 minutes) Learners return to the Driving Question Board started in Lesson 1. Each writes on a sticky note one NEW question or one answered question from today. Examples: 'Is the mean always a better summary than the mode?' 'What if one learner had 15 siblings?' 'Does the typical travel mode tell Mr Otieno what to build?' They place notes under the headings 'Answered' or 'New questions' and read 2-3 aloud.	Say: 'Write one question about typical values that you still have. Place it on the board.' WAIT TIME 10 seconds before the sharing. Cold-call 3 learners to read their notes. Move the question 'Which questions can the mode and mean help answer for Mr Otieno?' to the 'Answered' side, and mark questions about unusually large values and about the middle value as 'Coming in Lesson 4'. Ask: 'Which of today's ideas helps us answer the driving question?'	Driving Question Board tracking. A visible record of answered and new questions keeps the learning linked to the principal's decision and shows how each lesson adds to the evidence.	Check that notes are linked to today's content (mode, mean, typical) and that at least some are genuine questions (e.g., about unusual values), not statements. Count notes that identify limits of the mean or mode. Notice learners who have no note and invite them to add one verbally.

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Model Building Phase VIDEO: Estimating mean and median in data displays <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Calculating the mean Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(About 5 minutes) Learners open their draft model (the statistics cycle from Lesson 1: collect, organise, summarise, display, interpret) and add a 'Summarise' step with the new tools: Mode = most frequent value or category (works for categories and numbers); Mean = $\Sigma fx \div \Sigma f$ (numbers only). They add the evidence: Mode of travel = Walk (14 of 40, 35%); siblings mode 2, mean $83 \div 40 = 2.1$ siblings. They write one sentence on what these summaries still cannot tell Mr Otieno.	Say: 'Add the new step to your model. Show where the f column fits and write the formula in your own words.' WAIT TIME 10-15 seconds, then circulate and ask: 'Why does your model need both mode and mean?' Cold-call 2 learners to read their 'cannot tell' sentence (e.g., 'Mode: Walk does not show that most walk; the class is only 40 learners from one class'). Remind them that one class is a small sample and that Lesson 4 will ask which measure is best.	Cumulative model revision. Revising the class model after each lesson makes learners connect new skills (mode, mean) to the whole data cycle and the driving question.	Check that models include: (1) the mode defined correctly; (2) the mean formula $\Sigma fx \div \Sigma f$; (3) a note that the mean needs numbers; (4) travel and siblings evidence with exact numbers; (5) one stated limit. Use the exit ticket: 'Find the mean and mode of: x = 1, 2, 3 with f = 2, 5, 3.' Expected: $\Sigma f = 10$; $\Sigma fx = 2+10+9 = 21$; mean 2.1; mode 2.

D. TEACHER REFLECTION

Which learners confused the mode (2 siblings) with its frequency (12), or divided Σfx by the number of rows (6) instead of Σf (40)? Did the idea that the mean is a fair share (83 siblings shared among 40 learners) make sense to learners? Were any learners uncomfortable about sibling counts, and was the respectful norm kept? Did learners notice that Walk is the mode but only 35%, so 'almost everyone walks' is not supported? Which learners need more practice with an fx column before the median is introduced in Lesson 4?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The travel table has its highest frequency at Walk (14 of 40, 35%). The siblings table has its highest frequency at 2 siblings (12 of 40). Adding an fx column gives $\Sigma fx = 83$ for $\Sigma f = 40$.
What did I learn?	The mode is the value or category with the highest frequency and works for categories and numbers. The mean = $\Sigma fx \div \Sigma f$ and works only for numbers. Mean of siblings = $83 \div 40 = 2.075$, about 2.1 siblings; mode = 2. Travel mode: Walk is the mode but is not a majority.
How does this explain the phenomenon?	The typical Grade 10 Green learner walks to school and has about 2 siblings. The mean (2.1) need not be a whole number because it is a fair share. These summaries describe only this one class, so they are not enough alone to support a decision by Mr

	Otieno; we need to know when each measure is the best choice (Lesson 4).
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LESSON 4 (80 minutes (2 x 40-minute lessons)): Median and Choosing the Best Measure

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find the median of ordered ungrouped data for odd and even n, including from a frequency table using cumulative counting, and decide whether mean, median or mode best represents a data set, including the effect of an outlier.
Knowledge	By the end of the lesson the learner should be able to: (a) determine the median of ungrouped data (SLO 3.1 b: determine mean, mode and median of ungrouped data); (b) explain when the mean, median or mode best represents a data set, including how one extreme value affects the mean; (c) appreciate the importance of data organisation and interpretation in making effective decisions (SLO 3.1 e). Median = the middle value of ordered data: for odd n it is the $((n+1)/2)$ th value; for even n it is the mean of the two middle values.
Skills	Ordering data, cumulative counting from a frequency table, locating the middle position, calculating to 1 decimal place with units, comparing measures, and justifying a choice with evidence (NGSS-style practices: analysing and interpreting data; engaging in argument from evidence).
Attitudes	Core competencies: Critical thinking and problem solving (interpretation and inference when comparing measures). Values: Integrity (honest use of data, not removing a value just to get a nicer answer) and Love (considerate listening to peers' arguments). PCI: Life Skills (Decision Making), choosing the summary that gives a fair picture to decision-makers.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 3 gave the class the mode and the mean (siblings mean 2.1, mode 2; mode of travel = Walk). Lesson 4 adds the median and shows that the three measures can agree (siblings) or disagree (runner times), so the best summary depends on the data type and the question. This prepares learners to display data in Lessons 5 to 7 and to give Mr Otieno an honest recommendation in Lesson 8. Note: mean and median do not apply to travel mode because it is categories, so only the mode summarises it.
Safety Notes	Pen, paper and calculators only. Ensure learners do not tease peers about family size or about being slow runners. Use the fictional Eldoret club group and avoid naming real athletes or learners' real families; allow learners to opt out of sharing personal sibling data.

B. LESSON OVERVIEW

Mr Kamau returns to Mr Otieno's problem: the class has a table of travel data and can summarise number data, but which single number should the principal trust? Learners first predict which of three measures best describes a set of runner times. They then find the median of the siblings data using cumulative frequencies (20th and 21st values, both 2, so the median is 2) and the median of the Eldoret club 800 m times (4th of 7 values = 106 s), compare these with the mean and mode (109.3 s and 105 s), and test what happens to the mean when the 130 s time is removed (105.8 s). In the Explain phase they build a rule for choosing a measure. The class updates the Driving Question Board and revises its model of how raw data becomes evidence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall Lesson 3 (mode and mean). Mr Kamau writes the 7	Say: 'A selector in Eldoret asks the coach, how fast is this group? The	Prediction with commitment (Predict-Observe-Explain):	Check that at least 80% of learners have written a numerical

 VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Eldoret club 800 m times on the board in unordered form: 107, 130, 105, 104, 108, 105, 106 seconds. He tells learners the coach must tell a selector 'how fast this group is'. Individually, on paper, learners write: (1) a prediction of the 'typical' time, as a number, before any calculating; (2) which measure they think will describe the group best (mean, mode or something else) and why. They share with a partner for 2 minutes and then record agreements and disagreements. Learners also link back: 'In the travel data we could only use the mode. Will it be different for numbers?'	coach must give ONE number. Write down the number you would give before we calculate anything.' Give WAIT TIME of 10-15 seconds of silent thinking before any talk. Say: 'Now turn to your partner and compare. Do you agree?' Cold-call 4 pairs (not volunteers) to state their predicted number. Record all on the board without judging. Ask: 'Which of the measures we learned in Lesson 3, mean or mode, did you lean on, and why?' Do NOT mention the median yet. Circulate and note learners who pick the mean automatically.	learners commit to a number and a reason in writing before instruction. This surfaces the preconception that the mean is always the best summary and creates a cognitive need for comparing measures, which later evidence will test.	prediction AND a reason. Listen for: 'mean because it uses all the numbers' (common preconception) versus 'mode because most runners did 105'. Note the number of learners who predict between 104 and 110 s versus outside that range. Look for any learner who notices that 130 s looks unusual; record their name for use in the Explain phase.
Observe Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the median Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (siblings, same 40 learners): Learners use the siblings frequency table: 0 siblings 5; 1 sibling 9; 2 siblings 12; 3 siblings 8; 4 siblings 4; 5 siblings 2 (total 40). They add a cumulative frequency column: 5, 14, 26, 34, 38, 40. Since $n = 40$ is even, the median is the mean of the 20th and 21st values. In pairs they find both lie in the group with cumulative frequency reaching 26 (positions 15 to 26), so both values are 2 and the median is 2 siblings. They compare with the mean ($83/40 = 2.1$ siblings) and mode (2 siblings) from Lesson 3: all three agree. Part B (runner times): They order the 7 times: 104, 105, 105, 106, 107, 108, 130 s. $n = 7$ is odd, so the median is the $(7+1)/2 = 4$th value = 106 s. They find the mode, 105 s, and the mean, $765/7 = 109.3$ s. They then remove 130 s and find the mean of the other six: $635/6 = 105.8$ s. They record all results in an Observation Table: Data set Mean Median Mode Do they agree? Part C (even n	Distribute printed tables or have learners draw them. Say: 'First, ORDER the data. A median from an unordered list is wrong. Why must we order first?' Give WAIT TIME of 10-15 seconds. For siblings say: 'There are 40 learners. Which positions are in the middle? Count with me, 20th and 21st. Use the cumulative column to find which group the 20th learner is in.' Cold-call 3 learners to say which group contains the 20th and which contains the 21st. For runners say: 'Seven values, odd number. Use $(n+1)/2$. What position?' Cold-call 2 learners. Circulate with a checklist: Did they order? Did they use the right position? Correct common errors: finding the median as the middle of the list of categories (0 to 5, giving 2.5) rather than the middle of the 40 values; forgetting to divide by 7 for the mean. Say: 'Check your mean is not equal to any of the actual times. What do you notice about 109.3 compared with most of the times?' Remind	Structured data comparison table (Observation Table): learners place mean, median and mode side by side for three data sets so patterns across cases become visible. Cumulative counting links to Lesson 2's ordered frequency lists and makes the median accessible from a frequency table without writing out all 40 values.	Observable indicators: (1) learner orders the 7 times correctly before choosing the 4th value; (2) learner uses cumulative frequency to place the 20th and 21st values, writing 5, 14, 26, 34, 38, 40; (3) learner states the siblings median as 2 and the runner median as 106 s with units; (4) learner calculates $765/7 = 109.3$ s correctly to 1 d.p. Use a quick exit slip question mid-lesson: 'Median of 3, 5, 9, 12?' (answer 7). Record who gets 7 and who gives 5 or 9.

	practice): they find the median of the 6 times without 130 s: 104, 105, 105, 106, 107, 108, which gives the 3rd and 4th values 105 and 106, so the median is 105.5 s.	learners that median and mean must carry units: seconds, siblings.		
Explain Phase  VIDEO: Mean and standard deviation versus median and IQR <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the median: data displays Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4 learners discuss: (1) For the siblings data, why do mean, median and mode all give about 2? (2) For the runner times, why is the mean (109.3 s) larger than the median (106 s) and mode (105 s)? Which number would you give the selector? (3) What happened to the mean when the 130 s runner (who had a cold) was removed (109.3 s to 105.8 s)? What happened to the median (106 s to 105.5 s)? Which measure changed more? Groups draft a 'Choosing a Measure' rule such as: 'When the data has an extreme value (outlier), the median is more reliable than the mean; when the data is symmetrical with no outliers the three agree; for categories only the mode applies.' Learners then apply the rule to three quick cases and defend choices in writing: Mama Achieng's daily mandazi sales (40, 45, 42, 50, 65, 30, 20; total 292); the travel categories (Walk, Matatu, Boda boda, Bicycle, Private car, School bus); and the siblings table. They also link to the phenomenon: 'Why did Atieno say almost everyone walks when 14 of 40 is 35%? Which measure would the principal be given, and is it misleading?'	Pose: 'The 130 s runner had a cold. Should the coach delete that time to get a better number? Why or why not?' Give WAIT TIME of 10-15 seconds, then groups of 4 discuss for 5 minutes. Use roles: reporter, recorder, checker, timekeeper. Cold-call 3 groups (use random cards) to present one claim, one piece of evidence and one reasoning link (Claim-Evidence-Reasoning). Press: 'Which number in the table supports your claim?' Challenge integrity: 'If we remove 130 s just because it is inconvenient, is that honest?' Guide learners to the idea that an outlier should be reported, not hidden, and that the median is a more resistant summary. For the mandazi data say: 'Order the 7 sales: 20, 30, 40, 42, 45, 50, 65. What is the median?' (42 mandazi; the mean is $292/7 = 41.7$ mandazi). Ask: 'Here they are close. Why?' Link: 'For travel, can we find a median of Walk, Matatu, Boda boda? Why not?' Elicit that categories cannot be ordered numerically, so only the mode (Walk, 14 learners) applies. Ensure learners understand this about the class data only, not the whole school.	Claim-Evidence-Reasoning (CER) with argument from evidence: groups must state which measure best represents the runner data (claim), cite specific numbers (evidence: mean 109.3, median 106, mode 105, mean without outlier 105.8) and explain why one extreme value pulls the mean up (reasoning). Contrasting cases (siblings where measures agree versus runners where they differ) build a transferable rule rather than a memorised procedure.	Listen for correct reasoning: 'the 130 s is much bigger than the others, so it pulls the mean up but the median stays in the middle'. Success criteria: the group names the outlier, quotes at least two numbers, and states a measure with a reason. Look for the misconception that the outlier should be deleted without justification; note groups that argue honestly for reporting it. Collect a 3-line exit response: 'For the runner data I would give the selector ___ because ____.' Check at least 75% choose the median (106 s) or explain the mean with the outlier.
Driving Question Board (DQB) Creation  VIDEO: Interquartile range (IQR)	Learners return to the class Driving Question Board (started in Lesson 1) holding the question: 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' On sticky	Say: 'Look at the DQB. Which of our questions did today's evidence help to answer? Walk to the board and place a green dot on it.' Give WAIT TIME of 10-15 seconds before learners move. Cold-call 3 learners to read out a question	Driving Question Board as a living record of learning: learners see which questions are answered by evidence and which remain open, linking each lesson to the overall problem. Categorising questions helps them recognise the whole	Check that each learner has placed at least one new question or answer note. Look for questions that refer to evidence (numbers) rather than opinion. Note learners who connect median or outliers to Mr Otieno's decision, for example

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Effects of shifting, adding, & removing a data point Source: Khan Academy (English - US curriculum) Search ARES for similar readings	notes they (1) add new questions that arose, for example 'Which measure would Mr Otieno need if we collected the siblings or fares data from the whole school?' and 'Can a graph show the outlier better than a number?'; (2) move or tick off earlier questions answered today, such as 'Why do the mean and the mode tell different stories?'; (3) write one 'What we now know' note, for example 'Median is the middle of ordered data; an outlier pulls the mean but not the median'. They group questions into categories: Summaries (done), Graphs (coming in Lessons 5 to 7), Fair decision (Lesson 8).	they added and say which lesson might answer it. Ask: 'Which measure will Mr Otieno use for how learners travel?' (the mode, Walk, because the data are categories). Point out that the principal also needs a clear picture, which leads to graphs. Keep a visible list titled 'Questions for Lesson 5: how do graphs show this?'	statistics cycle (collect, organise, summarise, display, interpret).	whether one unusual class could mislead him. Record the number of new questions about displaying data (graphs) to confirm readiness for Lesson 5.
Model Building Phase  VIDEO: Median of Large Sets of Data Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise the class model, a flow diagram titled 'From messy list to fair decision', started in Lesson 1 and extended in Lessons 2 and 3. In their notebooks they add a new 'Summarise' box that now has three branches: Mode (most frequent; works for categories and numbers), Mean ($\Sigma fx \div \Sigma f$; uses every value; pulled by outliers) and Median (middle of ordered data; resists outliers). They add a decision arrow: 'Categories: use mode. Numbers with no extreme values: mean, median and mode are similar. Numbers with an outlier: report the median and mention the outlier.' They annotate the model with their own numbers: siblings (mean 2.1, median 2, mode 2) and runners (mean 109.3 s, median 106 s, mode 105 s, mean without 130 s 105.8 s). They finish with a 2-sentence note: 'What this means for Mr Otieno: ...'. They also write what the model still cannot do: show the pattern visually (to be solved with	Display the previous model. Say: 'Add today's learning. Where does the median fit? What decision arrow do we need?' Give WAIT TIME of 10-15 seconds before pair talk. Cold-call 3 learners to explain one branch of the model. Ask: 'Our class data cover only Grade 10 Green. What does that mean for how sure Mr Otieno can be?' Guide learners to the point that 40 learners is a small sample, so whole-school data are needed before deciding about the bus or lunch serving. Have learners complete the summary table prompts. Set a short follow-up task: 'Find the median of the 7 mandazi sales and say whether mean or median is better if Friday was a market day' (median 42; mean 41.7).	Cumulative model revision: learners extend an evolving diagram so that new ideas (median, outlier effect, choosing a measure) are linked to earlier ones (frequency table, mode, mean). Making the decision arrows explicit turns procedures into reasoning about which tool fits which data and question.	Check that the revised model includes all three measures with a correct condition for choosing each. Success criteria: correct definitions, correct numbers (median 2 siblings; 106 s; mean 109.3 s; 105.8 s without 130 s), and one statement linking to Mr Otieno or the limits of one class. Use a short rubric: Exceeds (explains outlier effect with numbers and sample limits), Meets (correct measures and choice rule), Approaches (correct calculations but no reasoning), Below (errors in the median position).

	graphs).			
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D. TEACHER REFLECTION

After the lesson ask yourself: Did learners order the data before finding the median, or did some pick the middle of an unordered list? How many used cumulative frequencies correctly for the 20th and 21st values? Did learners explain the outlier effect with numbers (109.3 versus 106 versus 105.8) or only say 'it is bigger'? Which learners still default to the mean as the 'best' measure? Did any learners tease peers about sibling numbers or running speed? What evidence shows learners understand that mean and median do not apply to travel categories? Plan to re-teach even-n medians (average of the two middle values) with 2 extra examples before Lesson 5 if more than a quarter of learners made errors.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We ordered the siblings data and found the 20th and 21st values both equal 2, so the median is 2 siblings (mean 2.1, mode 2). We ordered the 7 Eldoret club 800 m times and found the median 106 s, the mode 105 s and the mean 109.3 s; without the 130 s time the mean of the other six is 105.8 s.
What did I learn?	The median is the middle value of ordered data (for odd n the $((n+1)/2)$ th value; for even n the mean of the two middle values). One very large or small value (an outlier) pulls the mean but hardly moves the median. For categories such as travel mode only the mode (Walk, 14 of 40) can be used.
How does this explain the phenomenon?	The mean, median and mode agree for the siblings data because it is evenly spread, but they differ for the runner times because one slow time (130 s) pulls the mean up. So the best summary depends on the type of data and the question. For Mr Otieno this means choosing and explaining the measure honestly, and remembering that these data are from only one class.

LESSON 5 (80 minutes (double lesson of 2 x 40 minutes)): Bar graphs and line graphs: which picture fits which data?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners draw a bar graph for the travel frequency table and a line graph for Mama Achieng's daily mandazi sales, then decide which graph suits which type of data, so that the principal's evidence is shown in a form that can be read at a glance.
Knowledge	By the end of the lesson the learner should be able to: (a) represent data using bar graphs and line graphs; (b) interpret data from bar graphs and line graphs; (c) appreciate the importance of data organisation and interpretation in making effective decisions. Key ideas: a bar graph (equal-width bars with gaps, frequency axis starting at 0, both axes labelled, with a title) suits categories such as Walk, Matatu, Boda boda, Bicycle, Private car, School bus. A line graph (points joined by straight lines, time on the horizontal axis) suits change over time such as daily mandazi sales.
Skills	Choosing a sensible scale (0 to 14 or 16 in steps of 2 for travel; 0 to 70 in steps of 10 for mandazi); plotting bars and points accurately on squared paper; labelling axes and giving a title; reading values and trends from a graph; justifying the choice of graph with evidence. NGSS-style practices embedded: Developing and Using Models (a graph is a model of the data), Analyzing and Interpreting Data, and Engaging in Argument from Evidence.
Attitudes	Core competency: Critical thinking and problem solving (interpretation and inference when representing data). Value: Integrity (honest, accurate plotting while working in groups). Value: Love (considerate of others when extracting information and drawing conclusions from graphs). PCI: Life Skills (Decision Making): making informed decisions by identifying patterns in graphs and drawing conclusions.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 2 gave learners the travel frequency table (14, 12, 4, 3, 3, 4 = 40) and Lesson 4 showed that a summary should match the data. Lesson 5 turns the table into a bar graph so the class can see at a glance that Walk (14 of 40) is the largest group and Matatu (12 of 40) is second, so neither claim by Atieno or Brian shows a majority. It also introduces the line graph for data over time using the mandazi sales. The pie chart comes in Lesson 6 and critical reading of graphs in Lesson 7.
Safety Notes	Learners use rulers and pencils, so remind them to carry sharp pencils and rulers safely and not to throw them. Arrange desks so that groups can share squared paper without crowding. If phones or tablets are used to check graphs, follow the school device rules. Data are only about travel mode and mandazi sales and use fictional names, so there are no privacy concerns.

B. LESSON OVERVIEW

Mr Otieno still has money for only one of: a school bus route, a bicycle shed or a second lunch serving. The class has a frequency table, but a table of numbers is still hard to scan in a staff meeting. In this lesson Grade 10 Green at Lakeview Senior School, Kisumu, predicts which kind of picture would convince the principal fastest, draws a bar graph of the travel table (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; total 40), then draws a line graph of Mama Achieng's mandazi sales at Kibuye market (Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20; total 292). They compare the two graphs and build a rule for which graph fits which data. They add questions to the DQB and revise their model of how raw data becomes evidence. The lesson builds on Lesson 2 (frequency tables) and Lesson 4 (a representation should match the data) and does not teach pie charts, which come in Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Make bar graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown the travel frequency table from Lesson 2 (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; total 40) on the board, with a second table of Mama Achieng's mandazi sales beside it (Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20). Individually, in their books, they sketch in 3 minutes the picture they think would help Mr Otieno most for the travel table, and a picture they think would suit the mandazi sales. They write one sentence: 'I think this graph fits because...'. Then they pair up and compare. Their predictions connect to the phenomenon: which picture will settle the claims 'Almost everyone walks' and 'Most of us come by matatu'?	Say: 'Mr Otieno will read your work in two minutes before a meeting. Sketch the picture you think shows him the travel evidence fastest. Then sketch one for Mama Achieng's mandazi sales. No rubbing out: this is a prediction.' Give 3 minutes of silent sketching. Say: 'Turn to your partner and explain WHY you chose that picture.' After 2 minutes, cold-call 4 pairs (not volunteers) to show their sketches without judging. Ask: 'Did anyone draw the same picture for both tables? Why or why not?' WAIT TIME 10-15 seconds after each question before taking answers. Write the sketch types on the board (bars, line, others) without saying which is correct. Keep sketches visible for the Explain phase.	Predict-Observe-Explain with a Gallery of Initial Ideas: recording predictions before instruction makes learners' current thinking visible, so they notice later when the evidence confirms or changes it. Comparing sketches creates a real need to decide when to use bars and when to use a line.	Look for: (1) learners who sketch a picture for each table and give any reason; (2) whether they notice that travel mode is a list of categories while days follow each other in order; (3) misconceptions such as 'a line graph is always best because it looks neat' or 'bars without a starting zero are fine'. Record on a class list which learners chose the same type for both tables, and revisit these at the Explain phase.
Observe Phase  VIDEO: Creating a bar graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Line Graphs to Display Data Over Time (Flexbook) Source: CK-12  Search ARES for similar readings	Part A, bar graph (about 25 minutes). In groups of 4, learners use squared paper to draw a bar graph of the travel table. They choose a scale of 0 to 14 or 16 in steps of 2 on the frequency axis, start it at 0, draw 6 bars of equal width with equal gaps in the order Walk, Matatu, Boda boda, Bicycle, Private car, School bus, label both axes ('Means of travel', 'Number of learners') and write a title ('How Grade 10 Green travels to school, n = 40'). Each member plots 1 to 2 bars and checks a partner's bars against the table. They then read from their graph which bar is tallest (Walk, 14) and which two bars together reach 26 learners (Walk + Matatu = 65%). Part B, line graph (about 25 minutes). Learners draw a line graph of Mama Achieng's mandazi sales: days Mon to Sun	Distribute squared paper, rulers and pencils (one set per group of 4). Model only the first bar on the board (Walk at height 14) and then step back. Circulate and say: 'Show me where your frequency axis starts. Why does it start at 0?' and 'Tell me how you decided the width of your bars.' Cold-call 3 groups to explain how they chose their scale. For the line graph say: 'Which axis shows time? Why does time go along the bottom?' Pose the question: 'If we join the travel bars with a line, from Walk to Matatu to Boda boda, would the line mean anything?' WAIT TIME 10-15 seconds, then take 3 responses. Do not give the answer yet. Check drawings against the established data: travel bars 14, 12, 4, 3, 3, 4 and mandazi points 40, 45, 42, 50, 65, 30, 20. Ask	Constructing a model from evidence (Developing and Using Models): learners build each graph themselves from the table, so they see how every feature (starting at 0, equal bar widths, time on the horizontal axis) is a decision that affects how well the picture represents the data. Peer checking in groups builds accuracy and a shared language.	Observable indicators: frequency axis starts at 0 and uses equal steps; bars have equal width and gaps; all six bars match 14, 12, 4, 3, 3, 4; both axes and the title are present; line graph points match 40, 45, 42, 50, 65, 30, 20; learners can read Friday = 65 and Sunday = 20 and state that the bars total 40 and the sales total 292. Common errors to note: uneven scale, bars of different widths, leaving out axis labels, joining the travel bars with a line. Give a quick thumbs-up/sideways/down check on 'I can read my graph' before moving on.

	on the horizontal axis, mandazi sold on the vertical axis from 0 to 70 in steps of 10, points plotted for each day and joined with straight lines, with axes labelled and a title. They note the highest day (Friday, 65), the lowest day (Sunday, 20), and the rise from Thursday 50 to Friday 65 followed by the fall to Saturday 30 and Sunday 20. Each group checks that the total of the plotted values is 292.	groups finished early: 'What does Mama Achieng learn from your line graph that she cannot see quickly in the table?'		
Explain Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Line Graphs to Display Data Over Time Practice Source: CK-12  Search ARES for similar readings	Learners place their bar graph and line graph side by side and answer in groups: (1) What question does each graph answer? (2) Why would a line joining the travel bars be misleading? (3) Why do the mandazi points joined by a line make sense? (4) Which graph fits categorical data and which fits change over time? Groups write a class rule: 'Use a bar graph when... Use a line graph when...' with the travel and mandazi examples as evidence. Then they interpret the bar graph for Mr Otieno: Walk (14 of 40, 35%) is the tallest bar and Matatu (12 of 40, 30%) is second; neither is more than half, so 'almost everyone walks' and 'most of us come by matatu' are both wrong, though together they make 26 learners (65%). Bicycle is only 3 of 40 (7.5%) and School bus 4 of 40 (10%), so this class gives little support for a bicycle shed. Learners also note that the class bus count cannot confirm or refute Zawadi's remark that 'The bus is always full', because that is about the school's single 14-seater bus serving the whole school. They link back to Lesson 4: the picture must match the data, just as the best summary depends on the data.	Ask: 'Compare your two graphs. What is different about the data behind each one?' WAIT TIME 10-15 seconds, then cold-call 4 learners from different groups. Ask: 'Mama Achieng wants to know if sales are rising or falling during the week. Which graph tells her? How do you know?' Then: 'Atieno said almost everyone walks. Use the tallest bar to test her claim: what fraction of the class walks?' Cold-call 3 learners and write $14/40 = 35\%$ on the board. Ask: 'Is 35% almost everyone?' Press for evidence, not opinion: 'Point to the bar that shows it.' Ask: 'What does Zawadi's claim need that this class graph cannot give us?' Guide learners to the whole school and its single 14-seater bus. Keep the conclusion limited to Grade 10 Green. Compare their early predictions with what they have now drawn and ask: 'Whose prediction changed, and what changed it?' Do not teach pie charts: if a learner asks about circles, say 'Good question, we will park that on the DQB for Lesson 6.'	Claim-Evidence-Reasoning with Think-Pair-Share: learners make a claim about which graph fits which data and support it with the specific bars and points they drew, then reason from the type of data (categories versus time). Using the graph to test Atieno's and Brian's claims shows that a good representation turns a messy list into evidence.	Indicators: learners state that bar graphs suit categories (travel mode) and line graphs suit change over time (daily sales); they give a reason such as 'there is no order or in-between between Walk and Matatu, but Monday to Tuesday to Wednesday follows time'; they quote $14 \text{ of } 40 = 35\%$ and $12 \text{ of } 40 = 30\%$ from the bar heights; they say neither claim is more than half; they say the single class cannot settle the bus question. Use a quick exit slip: 'Name one data set for a bar graph and one for a line graph from your own life.' Mark acceptable examples (e.g. tree types at Lakeview, 30 trees: bar; a plant's height each week: line) and note learners who confuse the two.

Driving Question Board (DQB) Creation  VIDEO: Creating a bar graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Line Graphs for Data over Time: Strawberry Competition (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1: 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' Individually, they write one new question on a sticky note arising from today's graphs, for example 'Would the bars look the same for the whole school?', 'What if the bars started at 2 instead of 0?', 'Why can't we use a line for travel?' or 'How would a circle show the same travel data?'. They also move one earlier question to the 'partly answered' column with a note: for example 'Which picture shows the travel evidence best?' gets 'bar graph, from Lesson 5'. As a class they sort the new questions under headings such as 'Whole school data', 'Misleading graphs' and 'Other pictures'.	Say: 'Write one question that today's graphs made you wonder about. Put your name on the back.' Give 3 minutes of silent writing. Cold-call 5 learners to read their question aloud and place it on the DQB, asking each: 'Which part of the driving question does yours help with?' WAIT TIME 10-15 seconds before each response. Point out the link to Lesson 7: 'Questions about axes that start at the wrong number will be answered when we read graphs critically.' Point out the link to Lesson 6: 'The circle question will be answered when we draw pie charts.' Invite learners to move answered questions to the 'we think we know' area with one line of evidence from today. Ask one learner to summarise: 'What do we know now that we did not know at the start of this lesson?'	Driving Question Board as a living record of understanding: sorting questions into answered, partly answered and still open shows learners how their thinking has grown, gives them ownership of the investigation and sets a purpose for Lessons 6 and 7.	Indicators: questions refer to today's graphs and not just to earlier lessons; at least some questions go beyond the lesson, such as sample size, misleading axes or other picture types; learners can say which part of the driving question their question supports. Note learners who write only recall questions ('What is a bar graph?') and prompt them with 'What would you need to know to trust this graph?'
Model Building Phase  VIDEO: Mosaic plots and segmented bar charts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Create bar graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise the class model that started in Lesson 1, a flow diagram of how raw data becomes evidence for Mr Otieno. They add the new step: 'Display: choose the graph that fits the data type' with two branches and examples, 'Categories (Walk, Matatu, Boda boda, Bicycle, Private car, School bus) = bar graph, bars separate, frequency axis from 0' and 'Change over time (Mon to Sun mandazi sales) = line graph, time along the bottom, points joined'. Beside each branch they sketch a mini bar graph and a mini line graph and write one sentence of evidence from today, for example 'The tallest bar is Walk at 14 of 40, 35%' and 'Mandazi sales peak on Friday at 65 and drop to 20 on Sunday'. They also add a caution box: 'One class is a small sample,	Display the existing class model (raw list, tally, frequency table, mean/median/mode). Say: 'Add today's step. Where does a graph go in our model, and what decides which graph?' Give 5 minutes for groups to add it on poster paper. Cold-call 3 groups to present their added step and ask the others: 'Do you agree? What evidence from today supports it?' WAIT TIME 10-15 seconds. Ask: 'In our model, what still looks unfinished?' Guide learners to the pie chart and to checking graphs for tricks, then record both as 'To add' for Lessons 6 and 7. Collect the two-sentence notes to Mr Otieno as a written check. Finish with: 'Which would you trust more to show Mr Otieno the travel evidence, the raw list or the bar graph? Give me the evidence.'	Cumulative model revision: learners add one new component to a shared model and justify it with data from this lesson, so each lesson builds a coherent explanation of how raw data become fair evidence and the model is open to further revision.	Indicators: the model has a display step with two branches that match graph type to data type; examples use the correct data (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; mandazi 40, 45, 42, 50, 65, 30, 20); the note to Mr Otieno says the class gives little support for a bicycle shed (3 of 40, 7.5%) and that whole-school data are needed for the bus; learners do not claim the bus is full or empty from the class data. Note learners who still apply a line to categories and reteach with a quick question: 'Is there a value between Walk and Matatu?'

	so collect whole-school data before deciding on the bus or lunch serving'. Each learner writes a two-sentence note to Mr Otieno: what the bar graph says about a bicycle shed and what it cannot say about the bus.			
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners start the frequency axis at 0 and keep bars equal in width, or did some need reminding? (2) Could learners explain why a line does not make sense for travel categories, or did they only repeat the rule? (3) Which predictions from the Predict phase changed and why? (4) Did any learner use the bar graph to say that the class proves what the whole school does, or that Zawadi's claim about the bus is settled? Plan a quick correction if so. (5) Which learners need more practice with scales on squared paper before Lesson 6, where angles and a protractor are used, and before Lesson 7, where axes are examined critically? Keep a note of questions on the DQB that you can answer in those lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Bar graph of travel: Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4 (total 40), with the tallest bar for Walk. Line graph of Mama Achieng's mandazi sales: Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20, peaking on Friday and falling to Sunday.
What did I learn?	A bar graph suits categories, with separate equal-width bars and a frequency axis starting at 0. A line graph suits change over time, with time on the horizontal axis and points joined by straight lines. Both need labelled axes and a title.
How does this explain the phenomenon?	The same table of travel data becomes clear evidence in a bar graph: Walk is 14 of 40 (35%) and Matatu is 12 of 40 (30%), so neither claim shows a majority, though together they are 65%. Bicycle (3 of 40) gives little support for a shed. We should choose the graph that matches the data type, and collect whole-school data before deciding on the bus or lunch serving.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Pie charts: turning frequencies into angles

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners convert frequencies into angles using $f / n \times 360$, draw and label pie charts for the Grade 10 Green travel data and the Wanjiru family budget, and compare a pie chart with the bar graph of the same data to decide which representation best answers Mr Otieno's question.
Knowledge	By the end of the lesson the learner should be able to: (c) represent data using pie charts; (d) interpret data from pie charts. Learners will know that a pie chart shows each category as a share of a whole (the total frequency n), that the angles must add up to 360 degrees, and that a pie chart is best for 'how much of the whole?' while a bar graph is best for comparing exact counts.
Skills	Calculating angle = $f / n \times 360$ degrees (for travel data $360 / 40 = 9$ degrees per learner); using a protractor to draw sectors accurately; labelling sectors with category, frequency and percentage; checking that angles sum to 360; comparing representations of the same data; communicating a data-based claim. NGSS-style practices: analysing and interpreting data, using mathematics and computational thinking, constructing explanations, and engaging in argument from evidence.
Attitudes	Appreciate the importance of data organisation and interpretation in making effective decisions (KICD SLO e). Values: Integrity - learners are honest when measuring and recording angles; Love - learners are considerate of others when working with peers to extract information from charts and draw conclusions. Core competencies: Critical thinking and problem solving (interpretation and inference); Digital literacy (optional use of a phone or tablet spreadsheet to check the chart). PCI: Life Skills (Decision Making) - informed decisions from patterns in charts.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lesson 2 gave us the frequency table and percentages (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4 = 40). Lesson 5 gave us the bar graph of the same data. Now learners draw the pie chart, so they can see walk (126 degrees) and matatu (108 degrees) as shares of the whole class, and decide which graph helps the principal most. This moves the driving question forward: the same data in a different form can show a different thing clearly. Lesson 7 will then ask how to read graphs critically.
Safety Notes	Compasses and protractors have sharp points: carry them point-down, never point at others, and return them to the box. Keep the desk clear when drawing circles. If using phones or tablets offline, share devices fairly and keep bags off the floor walkways. Remind learners that the class data covers only Grade 10 Green and the names are fictional, so no learner should be teased about how they travel.

B. LESSON OVERVIEW

Mr Otieno still has money for only one of: a school bus route, a bicycle shed or a second lunch serving. In earlier lessons Grade 10 Green organised their 40 travel responses into a frequency table and drew a bar graph. In this lesson learners predict how the 360 degrees of a circle should be shared among the six travel groups, discover that each learner is worth $360 / 40 = 9$ degrees, and calculate and draw the travel pie chart (Walk 126, Matatu 108, Boda boda 36, Bicycle 27, Private car 27, School bus 36 = 360). They then apply the same method to the Wanjiru family monthly budget of KSh 30,000 in Nairobi (Food 144, Rent 90, School fees 54, Transport 36, Savings 36 = 360). Finally they compare the pie chart and bar graph of the travel data and decide when each is the better evidence. They see that walking and matatu together take 65% of the circle (234 degrees), neither is more than half, and the bicycle sector is small, which still gives little support for a shed from this one class.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Measuring angles in degrees Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pie Charts (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. Mr Kamau shows a blank circle on the board and says it stands for all 40 learners in Grade 10 Green. Without any calculation, each learner sketches on a sheet how they think the circle should be cut for Walk, Matatu, Boda boda, Bicycle, Private car and School bus, using the earlier claims (Atieno: 'Almost everyone walks'; Brian: 'Most of us come by matatu') as a guide. They write one sentence: 'I predict the biggest slice is ____ because ____.' They also predict: 'If the whole circle is 360 degrees, how many degrees might ONE learner be worth?' Groups share their sketches by holding them up. This connects to the phenomenon: the same 40 responses that looked meaningless as a raw list can now be seen as slices of a whole.	Draw a large circle and say: 'This whole circle is our whole class of 40. Before I give you any numbers, sketch how you think it should be shared. Do not calculate, just predict.' Give 3 minutes of silent individual work. WAIT TIME: after asking 'What do you think one learner is worth in degrees?', wait 10-15 seconds before taking answers. Cold-call 4 learners (2 who predicted walk, 1 who predicted matatu, 1 who predicted something else) with: 'Tell us your biggest slice and your reason.' Do not say who is correct. Record 3 or 4 predictions on the board under 'Our predictions'. Ask: 'Could one slice be more than half the circle? What would that mean in learners?' Collect the sketches to compare later.	Prediction with reasoning (individual first, then share): writing a prediction and a reason before instruction exposes prior ideas, such as 'the biggest slice must be more than half', and gives learners a personal stake in checking their prediction against the calculated chart later.	Observe whether learners (1) draw slices of different sizes rather than equal ones, (2) link slice size to the number of learners, and (3) give a reason, not just a guess. Note learners who think the circle has 40 degrees or 100 degrees. Tick on a class list who predicted walk, matatu or other. Listen for the idea of a 'whole' and its 'parts'.
Observe Phase  VIDEO: Pie Charts Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Pie Charts Practice Source: CK-12  Search ARES for similar readings	Groups receive the frequency table from Lesson 2 (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; total 40), a protractor, a ruler, a compass, plain paper and a calculation worksheet. Step 1: they find the angle for one learner, $360 / 40 = 9$ degrees. Step 2: they complete the table using angle = $f / n \times 360$: Walk $14 \times 9 = 126$, Matatu $12 \times 9 = 108$, Boda boda 36, Bicycle 27, Private car 27, School bus 36. Step 3: they add the angles and check the sum is 360 degrees. They also add a percentage column (35%, 30%, 10%, 7.5%, 7.5%, 10%) and see that percentage $\times 3.6$ gives the	Demonstrate one sector on the board: 'Watch my hands. I place the protractor centre on the circle centre, line up the zero line with my starting radius, and mark 126 degrees. Where do I begin the next sector?' Circulate and ask each group: 'How did you get 27 degrees for Bicycle? Show me the working.' Cold-call 3 groups to give the angle for a different category without looking at the answer line. WAIT TIME: after asking 'Why must all the angles add to 360?', wait 10-15 seconds before taking responses. Common errors to watch for: reading the wrong scale on the protractor (inner versus outer), starting each new sector	Stepwise worked procedure with peer checking: learners connect the proportion idea (14 out of 40) to the angle (14 out of 40 of 360) through a calculation table, then check one another's constructions. Using the total-equals-360 check builds the idea that the pie chart is a whole divided into parts, linking back to the percentages from Lesson 2.	Check (1) angles calculated correctly using $f / n \times 360$, (2) the sum of angles equals 360, (3) sectors drawn within 2 degrees of the true value, (4) sectors in the correct order, (5) title and labels present. Use a quick checklist: Calculation / Accuracy / Labels. Note groups who use 9 degrees per learner as a shortcut and ask them to explain why it works only for $n = 40$.

	same angle. Step 4: they draw a circle of radius 5 cm, mark a starting radius, and use a protractor to draw each sector in the order Walk, Matatu, Boda boda, Bicycle, Private car, School bus. Step 5: they label each sector with the category, frequency and percentage, and give the chart a title. Groups swap charts and check each other's angles with a protractor. Early finishers repeat for the Wanjiru budget angles (144, 90, 54, 36, 36).	from the circle edge rather than the last radius, and angles that sum to 359 or 361 because of rounding. Say: 'If your total is not 360, find the mistake before you draw.' Remind: 'Sharp points down, please.'		
Explain Phase  VIDEO: Measuring angles in degrees Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pie Charts: Soda Party (PLIX) Source: CK-12  Search ARES for similar readings	Learners lay their travel pie chart next to the bar graph of the same data from Lesson 5. In groups they complete a comparison table with three columns: 'Easier to see on the pie chart', 'Easier to see on the bar graph' and 'Same on both'. They answer: (1) Is the Walk slice more than half the circle? (126 is less than 180). (2) Which two sectors together are almost two-thirds of the circle? (Walk + Matatu = 234 degrees = 65%). (3) What can the bar graph show that the pie chart cannot show easily? (exact counts, and comparing Bicycle 3 with Private car 3). They then repeat the reasoning for the Wanjiru family budget pie chart: Food takes 144 degrees (40%), more than Rent (90 degrees, 25%). In a class talk they write a two-sentence claim for Mr Otieno: 'On this evidence from Grade 10 Green, _____. But because _____, we should _____.' They revisit Atieno's and Brian's claims and decide, using angles, that neither is more than half. They note that Zawadi's remark about the bus cannot be tested with this class data.	Pose: 'Atieno said almost everyone walks. Use your pie chart to give evidence for or against.' Give 1 minute silent think time, then 10-15 seconds WAIT TIME after asking 'What fraction of the circle would almost everyone be?' Cold-call 4 learners with different views and use the stems: 'I notice _____', 'The angle shows _____', 'I disagree because _____'. Ask: 'If the principal only had 10 seconds, which graph would you hand him? Why?' Press for reasoning with 'How do you know?' and 'What would change your mind?'. Probe sampling: 'Does this class speak for the whole school?' Guide towards the recommendation: no money for a shed on this evidence, and collect whole-school data before deciding on the bus or lunch serving. Do not teach misleading axes yet; that is for Lesson 7.	Compare-and-contrast of representations plus claim-evidence-reasoning: learners place two displays of the same data side by side to see that each shows something different, then write a claim supported by angle and percentage evidence. This builds the understanding that the best representation depends on the question being asked.	Look for learners who (1) use angles or percentages as evidence rather than just 'it looks big', (2) correctly say that no sector is over 180 degrees so no travel mode is more than half, (3) state one advantage of each graph, and (4) mention that one class is a small sample. Collect the two-sentence claims as an exit slip and sort them into 'evidence-based', 'partly' and 'opinion only'.
Driving	Learners return to the class	Say: 'Look at your pie chart. What	Driving Question Board	Check that questions are linked to

Question Board (DQB) Creation  VIDEO: Pie Charts Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Measure angles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Driving Question Board headed 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' On sticky notes they add new questions that came from the pie chart, such as 'Would the pie chart look the same for the whole school?', 'Can a pie chart be used for the runner times or the siblings data?', 'What if two sectors are almost equal?', and 'How could someone make a pie chart look unfair?'. They place their notes under headings such as 'Collecting', 'Organising', 'Summarising', 'Displaying' and 'Deciding'. They also move one earlier question, such as 'How can we see who is right?', to an 'Answered' area and attach their pie chart as evidence.	does it let us answer now? What new question does it make you ask?' Give 2 minutes silent writing, then 10-15 seconds WAIT TIME before anyone shares. Cold-call 3 learners to read a note aloud and say where it goes on the board. Ask: 'Which of these questions can we answer next lesson?' Link: 'The question about unfair graphs, we will come back to in Lesson 7.' Photograph or record the DQB for the next lesson.	questioning: learners turn what they have just learned into new questions and sort them, which keeps the inquiry connected to the principal's decision and shows how each lesson adds to the evidence.	the pie chart or the decision (not unrelated), that learners can say which step of the statistics cycle the question belongs to, and that at least some questions go beyond simple recall (for example about fairness or sample size). Note which learners are ready to move an earlier question to 'Answered'.
Model Building Phase  VIDEO: Area of a sector <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Draw angles Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open their class model, started in Lesson 1 and revised since, on how to turn a messy list into evidence. In groups they add a new section called 'Display as a share of the whole: pie chart' with: (1) the formula $\text{angle} = f / n \times 360$, (2) the check that angles sum to 360, (3) a small labelled travel pie chart (Walk 126, Matatu 108, Boda boda 36, Bicycle 27, Private car 27, School bus 36), and (4) a note on when a pie chart is a good choice (parts of a whole, few categories) and when a bar graph is better (exact counts, comparing close values). They also add a short rule of thumb: 'Pie chart for shares of a total such as the Wanjiru family KSh 30,000 budget; bar graph for comparing counts.' Then they complete the summary table in their books, revising their earlier claim to Mr Otieno.	Say: 'Open your model. What new piece does the pie chart add to how we turn messy data into evidence?' Ask each group to explain one new arrow or box. WAIT TIME 10-15 seconds after 'Where in our model does the total 40 matter?' Cold-call 3 groups to share their rule of thumb and challenge it: 'Would a pie chart work for 15 different tree types? Why or why not?' Prompt revision: 'Has anything in your first model changed because of what you drew today?' Remind them the whole school must be surveyed before a final decision on the bus or lunch serving. Collect the models and the summary tables.	Cumulative model revision: learners add the pie chart step to their evolving model of the statistics cycle and explain how it connects to frequency tables, bar graphs and the principal's decision. This consolidates the idea that each step transforms the data into clearer evidence.	Look for models that (1) include the formula $f / n \times 360$ and the 360 check, (2) link the pie chart to the frequency table and total n, (3) state when a pie chart or bar graph is better, and (4) connect to the decision for Mr Otieno. Use a 3-level check: Not yet (formula only), Developing (formula plus drawing), Secure (formula, drawing and reasoned choice of representation).

D. TEACHER REFLECTION

After the lesson, consider: Did learners explain WHY $\text{angle} = f / n \times 360$ works, or did they only follow the steps? How many struggled with the protractor and how could that be supported next time (for example, a cut-out 9-degree wedge)? Did the pie chart versus bar graph comparison lead to reasoned choices rather than a simple preference? Did anyone notice that a slice below 180 degrees means less than half? Which learners used percentages and angles as evidence in their claims? Were any learners uncomfortable about how they travel to school, and how can the discussion remain respectful? Note questions on the DQB that need to be picked up in Lesson 7, especially fairness of graphs, and plan how to use them.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the 40 travel responses can be shown as a pie chart: Walk 126 degrees, Matatu 108 degrees, Boda boda 36 degrees, Bicycle 27 degrees, Private car 27 degrees and School bus 36 degrees, which add up to 360 degrees. Walk takes 14 of 40 learners (35%) and Matatu 12 of 40 (30%), so together 65%, but neither is more than half the circle. We also drew the Wanjiru family budget of KSh 30,000 (Food 144, Rent 90, School fees 54, Transport 36, Savings 36 = 360).
What did I learn?	We learned that $\text{angle} = f / n \times 360$, that each learner in our class is worth 9 degrees, that the angles of a pie chart must add up to 360, and that a pie chart shows each part as a share of the whole while a bar graph shows exact counts and is better for comparing close values.
How does this explain the phenomenon?	We can explain that the pie chart helps Mr Otieno see at a glance that walking and matatu are the biggest groups but neither is the majority, and that only 3 of 40 (7.5%) cycle, so this class gives little support for a bicycle shed. Because this is only one class, we advise collecting the same data from the whole school before deciding on the bus or second lunch serving.

LESSON 7 (80 minutes (2 x 40-minute lessons)): Reading Graphs Critically and Using Digital Tools

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners read bar graphs, line graphs and pie charts to extract information and defend conclusions with numbers, spot a misleading graph whose vertical axis starts at 2 instead of 0, and recreate a graph using a spreadsheet or graphing app, so they can give Mr Otieno trustworthy evidence.
Knowledge	By the end of the lesson the learner should be able to: (d) interpret data from bar graphs, line graphs and pie charts; (c) represent data using bar graphs, line graphs and pie charts on a digital device; (e) appreciate the importance of data organisation and interpretation in making effective decisions. Learners know that a bar graph's frequency axis must start at 0, and that a truncated axis makes walk (14) look 12 times bicycle (3) when the true ratio is 14:3, about 4.7 times.
Skills	Extracting information from graphs and charts, drawing and defending conclusions with numbers, comparing a graph with its frequency table to detect distortion, and entering data and creating labelled graphs on a spreadsheet or app (SEP: Analysing and interpreting data; Engaging in argument from evidence; Using mathematics and computational thinking).
Attitudes	Core competencies: Critical thinking and problem solving (interpretation and inference when reading graphs); Digital literacy (using a digital device to draw graphs and charts). Values: Integrity (honest graphs, truthful scales); Love (considerate when working with peers to extract information and draw conclusions). PCI: Life Skills (Decision Making), making informed decisions by identifying patterns in graphs and drawing conclusions.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lessons 5 and 6 taught learners to draw the bar graph, line graph and pie chart. Lesson 4 taught them that the best measure depends on the data. In Lesson 7 they stop drawing and start judging: can a graph be trusted, and what exactly does it let us claim? This prepares the final recommendation to Mr Otieno in Lesson 8. Walk 14 (35%) and Matatu 12 (30%) together make 65%. Bicycle is 3 (7.5%) and School bus 4 (10%). Only one class is covered, so conclusions must be limited.
Safety Notes	Handle phones, tablets and computers carefully, share devices fairly and keep chargers away from water. Work offline with the installed spreadsheet or graphing app; do not browse the internet. Keep bags and cables out of walkways. Learners who use a phone should not share personal information or photographs. Rulers and protractors are used with care.

B. LESSON OVERVIEW

Mr Kamau shows Grade 10 Green of Lakeview Senior School, Kisumu, two bar graphs of the same travel data, one with the axis starting at 0 and one starting at 2. Learners predict which one tells the truth about walking and cycling, then, working in pairs, examine four graphs: the honest travel bar graph, the misleading one (walk 12 units, bicycle 1 unit, apparently 12 times, truly 14:3), the travel pie chart (angles 126, 108, 36, 27, 27, 36 = 360) and Mama Achieng's mandazi line graph (Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20, total 292). Pairs extract facts, write one conclusion each with a number to back it up, and say what a graph cannot show, such as Zawadi's claim about the school's single 14-seater bus, which a count of 4 learners from one class cannot confirm or refute. They then recreate the travel bar graph and pie chart on a spreadsheet or app and test what happens when the axis start is changed. The lesson ends by adding to the Driving Question Board and revising the class model of how messy data becomes fair evidence for Mr Otieno.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reading bar charts: comparing two sets of data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Create picture graphs (picture more than 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Mr Kamau projects or draws two bar graphs of the same Grade 10 Green travel data side by side without the frequency table. Graph A has the vertical axis starting at 0. Graph B has it starting at 2. Individually, learners write for 2 minutes: 'Which graph tells the truth about how many more learners walk than cycle? How many times taller is the Walk bar than the Bicycle bar in each graph?' They sketch a prediction and then compare with a partner. They are not yet told the answer. Link: the principal may see only one graph, so which one he sees may change his decision.	Say: 'Mr Otieno has a busy day and will look at only one of these graphs for ten seconds. Study both quietly and decide which one he should trust.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and compare for one minute.' Cold-call 4 pairs for their prediction and reason, without confirming or correcting. Ask: 'What would you check to be sure?' WAIT TIME: 10-15 seconds after each question before calling on anyone. Record the predictions in a tally on the board (A is true, B is true, both are true, cannot tell). Say: 'Keep your prediction, we will test it with the evidence.'	Predict-before-instruction with a Notice and Wonder frame. Committing to a written prediction exposes the common idea that a taller bar always means a bigger number, creating a need to check the scale. Recording the vote publicly gives the class a claim to revisit, which builds the habit of testing a graph instead of trusting the picture.	Look for: (1) learners who write a tallest-bar ratio rather than only 'it looks bigger'; (2) learners who mention the starting number on the axis, or ask for the table; (3) learners with the misconception that both graphs are equally true because 'the bars are there'. Tally the votes and note the 3 pairs who raise the axis, as evidence for who to call on later.
Observe Phase  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve problems with bar graphs 1 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Pairs receive a Graph Detective sheet with four displays from Grade 10 Green and Mama Achieng's data: (1) the honest travel bar graph (axis 0 to 16, steps of 2); (2) the misleading bar graph (axis starting at 2); (3) the travel pie chart; (4) the mandazi line graph for Mon to Sun. For each, the pair completes: 'What does the title and each axis tell us? Read three exact values. Write one conclusion with a number.' On Graph 1 they read Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4. On Graph 2 they measure bar heights: Walk 12 units, Bicycle 1 unit, so apparently 12 times, and compare with the true 14 to 3, about 4.7 times ($14 \div 3 = 4.67$). On Graph 3 they use the angles to	Distribute the sheets and say: 'You are graph detectives. Every claim needs a number from the graph.' Circulate and ask each pair: 'Show me where on the graph you got that value.' At Graph 2, ask: 'Measure the Walk and Bicycle bars. How many times taller is Walk? Now divide the real frequencies, 14 by 3. Do the answers agree?' Point learners who stall to the axis start. For the pie chart, ask: 'How can you use degrees to check the percentage? What fraction of 360 is 126?' For the line graph, ask: 'Which two days show the largest change? What might explain it at Kibuye market?' For Zawadi's bus claim, ask: 'Can this graph prove the bus is always full? What other data would we need?' Use WAIT TIME	Evidence-based reading with Claim-Evidence-Limit sentence frames: 'I claim ... because the graph shows ... but the graph cannot tell us ...'. Cross-checking one graph against another (pie against bar, bar against table) lets learners find the distortion themselves and builds argument from evidence about what a graph can and cannot support.	Observable indicators: (1) learners read exact values rather than estimates, for example Walk 14 and Bicycle 3; (2) they compute $14 \div 3 \approx 4.7$ and compare with 12; (3) they convert 126 degrees to 35% correctly; (4) each conclusion includes a number; (5) they name a limit such as 'one class only' or 'cannot say about the bus'. Check sheets for pairs who give conclusions without numbers, and note which pairs misread the axis scale.

	check: $126 \div 360 = 35\%$ for Walk and $126 + 108 = 234$ degrees, which is 65% for Walk plus Matatu. On Graph 4 they find Friday highest (65), Sunday lowest (20), total 292, and the rise from Thursday 50 to Friday 65 and the fall to Saturday 30. They also write one question each graph cannot answer, for example 'Is the school bus always full?', since only 4 of the 40 in this class use it and the school has a single 14-seater bus serving the whole school.	of 10-15 seconds before offering hints. Give 20 minutes. Stop and cold-call 5 pairs, each reading one value and one conclusion.		
Explain Phase  VIDEO: Reading bar charts: comparing two sets of data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Read bar graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Whole-class discussion. Pairs share what they found and agree on a rule: how to spot a misleading graph. They list checks such as: does the frequency axis start at 0? Are the steps equal? Are both axes and the title labelled? Are bars the same width? Do the pie angles add to 360? Then they defend conclusions for Mr Otieno using the honest graphs: walking is the largest group (14 of 40, 35%) and matatu second (12 of 40, 30%); 'almost everyone walks' and 'most come by matatu' are both wrong because neither is more than half, though together they are 26 learners, 65%. Only 3 of 40 (7.5%) cycle, so this class gives little support for a bicycle shed. Learners connect to Lesson 4: for travel categories only the mode (Walk, 14) applies, since mean and median do not apply to categories. They also discuss which display suits which question: the pie chart for parts of the whole, the bar graph for comparing categories, the line graph for change over time. Digital part: pairs open a spreadsheet or graphing app on a shared device, type the travel frequency table	Say: 'Tell me one check that stops us being fooled by a graph.' Cold-call 6 learners, one check each, and list them on the board. WAIT TIME: 10-15 seconds. Ask: 'Why might someone draw the axis from 2? Was it dishonest or just careless? What should an honest data person do?' Link to Integrity. Ask: 'Atieno said almost everyone walks. Use the graph to say whether she is right, with a number.' Cold-call 3 pairs. Ask: 'Can we tell Mr Otieno to build a bicycle shed?' and 'Can one class decide the bus or lunch serving?' Guide to the conclusion that the data covers only Grade 10 Green and the whole school must be surveyed. For the digital task, model the first step only: entering the table, then circulate. Say: 'Change the axis minimum to 2 and watch Walk and Bicycle. What changed, and what did not change in the data?' Share devices so every learner types or reads at least once. Remind them: no internet needed.	Argument from evidence, with a Consensus Discussion and a class-built checklist. The checklist turns noticing into a reusable criterion for critical reading. Changing the axis on the digital tool is a manipulation experiment: the data stays fixed, only the display changes, so learners see that the picture, not the numbers, is what is misleading. This builds their understanding of which summary and graph to trust.	Indicators: (1) learners state at least three checks, including axis starting at 0; (2) they use numbers (14, 12, 35%, 65%, 7.5%) in defending conclusions; (3) they say the data covers one class only; (4) on the device, the graph has a title, labelled axes and correct values; (5) they explain that changing the axis changes the look but not the data. Use a 3-question exit ticket: 'Why should a bar graph start at 0? Walk 14, Bicycle 3: what is the true ratio? What would you tell Mr Otieno about a bicycle shed?' Look for 14:3 or about 4.7 times.

	(Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4), create a bar graph and a pie chart, add a title and axis labels, then change the axis minimum from 0 to 2 and observe how the bars change. They check that the pie chart percentages match 35%, 30%, 10%, 7.5%, 7.5%, 10%.			
Driving Question Board (DQB) Creation  VIDEO: Reading line graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Read bar graphs (2-step problems) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board: 'How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?' In pairs they write on sticky notes: (1) one answered question, for example 'Why did the second graph make walking look so much bigger? Because the axis started at 2'; (2) one new question, for example 'How many learners must we ask in the whole school before we trust the results?' or 'How should we report that the bus claim cannot be checked from our class?'. They place the notes on the board under headings such as Reading Graphs, Honest Graphs, Digital Tools and Still Unsure.	Say: 'Write one thing you can now answer and one thing you still want to know.' Give 3 minutes. Cold-call 4 pairs to read a note aloud and place it. WAIT TIME: 10-15 seconds. Ask: 'Which question will be most important for the final recommendation to Mr Otieno?' Point out that Lesson 8 will pull all the evidence together. Move answered notes to an Answered area and keep open questions visible. Praise notes that carry a number or a limit.	Question tracking using the DQB. Sorting notes into answered and open questions makes learning visible and helps learners see how Lesson 7 moves them from making graphs to judging evidence, and which gaps remain before they can advise the principal.	Indicators: (1) at least 70% of pairs post one answered and one new question; (2) new questions mention the sample size, the type of graph, or fairness; (3) answered notes use correct terms such as axis, scale, frequency. Note any pairs whose questions still show confusion about the axis, for revisiting at the start of Lesson 8.
Model Building Phase  VIDEO: Reading bar graphs: Harry Potter <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners open the class model of how messy data becomes fair evidence: raw list of 40 responses, then tally and frequency table (Lesson 2), then summaries (modes, means, medians in Lessons 3 and 4), then displays (bar, line and pie in Lessons 5 and 6). They add a new final stage: 'Check the display is honest and read it carefully'. In pairs they add a Graph Checklist box (axis starts at 0, equal steps, title, labels,	Say: 'Our model has a new stage today. Where does it go and why?' Give pairs 5 minutes to revise the model, then cold-call 3 pairs to explain a change. WAIT TIME: 10-15 seconds. Ask: 'If Mr Otieno saw the misleading graph first, which decision might he make and what would that cost the school?' Ask: 'Which display would you give him for the travel data and why?' Expect: bar graph or pie chart, with the table beside it. Check that	Cumulative model revision. Adding a critical-reading stage to the model helps learners integrate organising, summarising, displaying and judging into one process, and prepares them to build their final recommendation in Lesson 8. Writing a short explanatory note forces them to connect the model to the evidence and the principal's decision.	Indicators: (1) the revised model includes a checking-the-graph stage and a limits arrow; (2) the 2-sentence note states that the axis must start at 0 and that the bicycle shed is not supported by this class data (3 of 40, 7.5%); (3) numbers in the model match the anchor data; (4) learners say the whole school should be surveyed. Collect models and use the checklist to decide who needs extra support before the Lesson 8 project.

 EXERCISE: Create picture graphs (picture more than 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	angles sum to 360, claim has a number, state the limit) and a 'Limits' arrow showing that one class cannot speak for the whole school. They revise their draft model and write a 2-sentence note: 'The principal can trust a graph if For our data, he should not ... because' They add to the summary table what was observed, learned and explained.	models keep the numbers exact: 14, 12, 4, 3, 3, 4 = 40; 35%, 30%, 65%. Encourage learners to note a gap for Lesson 8: the whole-school data needed before choosing the bus or lunch serving.		
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners find the axis problem themselves by comparing the 12 units to 1 unit with the true 14 to 3, or did they need me to point it out? Which pairs wrote conclusions with numbers and limits, and which only described the picture? Were devices shared fairly, and did every learner enter or read data at the spreadsheet? Which learners still believe a taller bar always means a bigger number? What will I revisit at the start of Lesson 8 so every pair can make a fair, data-based recommendation to Mr Otieno?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the same travel data looked very different on two bar graphs: with the axis starting at 0, Walk (14) is about 4.7 times Bicycle (3); with the axis starting at 2, Walk looked 12 times Bicycle. We also read the pie chart and the mandazi line graph, and recreated the travel graphs on a spreadsheet or app.
What did I learn?	We learned to read exact values from bar graphs, pie charts and line graphs, to support every conclusion with a number, and to check a graph for a frequency axis that starts at 0, equal steps, labels, a title and pie angles that add to 360. We also learned that one class cannot speak for the whole school.
How does this explain the phenomenon?	A misleading graph changes how the data looks, not the data itself. Honest graphs of the travel data show Walk 14 of 40 (35%) and Matatu 12 of 40 (30%), together 65%, and only 3 of 40 (7.5%) cycling, so the class gives little support for a bicycle shed. Mr Otieno should trust a graph only after checking its axis and scale, and should collect whole-school data before deciding on the bus or lunch serving.

LESSON 8 (80 minutes (2 x 40-minute lessons)): Project: Evidence for the Principal

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To synthesise the whole statistics cycle (collect, organise, summarise, display, interpret) into a data-based recommendation for Mr Otieno and to answer the driving question. No new technique is taught.
Knowledge	By the end of the sub strand, the learner should be able to: a) draw a frequency distribution table for ungrouped data; b) determine mean, mode and median of ungrouped data; c) represent data using bar graphs, line graphs and pie charts; d) interpret data from bar graphs, line graphs and pie charts; e) appreciate the importance of data organization and interpretation in making effective decisions.
Skills	Communication and collaboration; Critical thinking and problem solving: interpretation and inference when drawing conclusions from the Grade 10 Green travel data. Digital literacy: using a spreadsheet or graphing app (offline-capable) to display data. Science and Engineering Practices: analysing and interpreting data; constructing explanations; engaging in argument from evidence; developing and using models.
Attitudes	Integrity: honest reporting of what the data does and does not show. Love: consideration of others when peers present and critique. Life Skills (Decision Making): making informed decisions from patterns in graphs and charts.
Key Inquiry Question	How do we use statistics in day-to-day life?
Purpose in Storyline	Lessons 1 to 7 moved from an unsettled raw list (Lesson 1), to frequency tables (2), mean, mode and median (3-4), bar, line and pie graphs (5-6), and critical reading of graphs and digital tools (7). In Lesson 8 learners bring every step together, present recommendations to Mr Otieno, compare the Lesson 1 draft model with the final model, and complete the DQB to answer: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?
Safety Notes	Learners use only their own anonymous class travel data; no names are linked to responses. Treat all peer presentations with respect and critique the evidence, not the person. Phones or tablets: use offline, keep volume low, and follow school device rules. Protractors and compasses: carry with points down.

B. LESSON OVERVIEW

Lakeview Senior School in Kisumu: Mr Otieno has money for only one of a school bus route, a bicycle shed or a second lunch serving. In Lesson 1 Atieno said 'Almost everyone walks', Brian said 'Most of us come by matatu', and Zawadi said 'The bus is always full', and the raw list of 40 responses settled nothing. Today Grade 10 Green finishes the project. Groups run the full cycle on the anchor data (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; total 40) and present an evidence-based recommendation: walking is the largest group (35%), matatu second (30%), together 65%; neither is more than half, so both claims are wrong; bicycle is only 7.5% and school bus 10%, so no money for a shed on this evidence; and because one class is a small sample, collect the same data from the whole school before deciding on the bus or lunch serving. Zawadi's claim concerns the single 14-seater bus serving the whole school, so 4 learners on the bus in this class can neither confirm nor refute it. Learners also use siblings and runner times to explain why the best summary depends on the data type. They then compare the Lesson 1 draft model with a final model, complete the DQB and write the Final Explanation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Lesson 1 when the list of 40 responses could not settle the claims of Atieno, Brian and Zawadi. On a sticky note or in their books they write a prediction: 'If I give Mr Otieno ONE page of evidence, what should be on it, and what should his decision be about the bicycle shed?' They also predict which claim, 'Almost everyone walks' or 'Most of us come by matatu', will survive the full analysis. They privately rate their confidence from 1 to 5. They share with a partner for 2 minutes, then with the class.	Display the Lesson 1 draft model and the DQB on the board. Say: 'Three weeks ago we could not tell who was right. Today you are the statisticians advising Mr Otieno. Before we open any tables, write your prediction.' Give WAIT TIME of 15 seconds of silence. Cold-call 4 learners for predictions: 'Wanjiru, what would you put on the one page? Why?' Record predictions without judging. Ask: 'What would make Mr Otieno trust your page?' Do NOT reveal the totals yet; keep Walk 14, Matatu 12 and the rest hidden until learners retrieve them from their own work in the Observe Phase.	Prediction with confidence rating and Think-Pair-Share: surfaces prior ideas about how data becomes evidence, so learners can compare their starting ideas with their final model and see their own growth.	Look for: predictions that mention organising data (table), a display (graph or chart) and a summary, not just 'count them'; learners who name 'whole school' or 'more data'; learners who still predict one claim is simply right. Tally how many name each statistics step; reteach any step missing in the groups' later work.
Observe Phase  VIDEO: Mean, median, & mode example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Make bar graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4 (roles: data manager, table maker, grapher, spokesperson) learners rebuild the evidence pack from their own Lesson 2 to 7 work. (1) Tally the raw list (Learners 1-40) into a frequency table and check the frequencies add to 40: Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4. (2) Compute percentages ($f/n \times 100$: 35%, 30%, 10%, 7.5%, 7.5%, 10%, sum 100%) and note Walk + Matatu = 26 learners = 65%. (3) State the mode of travel (Walk) and explain why mean and median do not apply to categories. (4) Draw a bar graph (frequency axis from 0, steps of 2, bars equal width with gaps, both axes labelled, title) and a pie chart (9 degrees per learner: 126, 108, 36, 27, 27, 36 = 360). (5) Use the siblings table (mean 83/40 = 2.1, median 2, mode 2) and the Eldoret runner times (mean 109.3 s, median 106 s, mode 105 s; mean	Circulate with a checklist. Ask: 'Show me where your frequencies add to 40.' 'Which angle did you get for Matatu, and how did you get 108?' 'Why can you not find a mean of travel mode?' Give WAIT TIME of 10 seconds after each question before prompting. Cold-call 3 groups to state their totals check. For stuck groups, point back to the earlier lesson's method without re-teaching: 'Look at your Lesson 2 table, what did we do with the tally marks?' Remind: 'Write mean, never average, and give units: seconds, learners.' Check that no learner names are linked to responses.	Evidence-pack building (data-to-evidence chain): learners physically order the cycle collect, organise, summarise, display, interpret, so they see that each step makes the data more usable than the raw list. Comparing tables, graphs and summaries for the same data shows why representation matters.	Observable indicators: frequencies total 40 and percentages total 100%; pie angles total 360; bar graph axis starts at 0 with labels and title; learner states mode = Walk and explains that mean and median do not apply to categories; learner explains that the runner mean (109.3 s) is pulled up by the 130 s outlier while the median (106 s) represents the group better. Use a 4-item checklist per group: table, summary, graph, caution.

	105.8 s without the 130 s time) to show when mean, median and mode agree or differ. (6) Check a truncated-axis version of the bar graph (starting at 2) and calculate the true walk to bicycle ratio of 14:3, about 4.7 times, not 12 times. Groups may use a spreadsheet or app offline for the bar graph or pie chart.			
Explain Phase  VIDEO: Mean, median, & mode example - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Bar Graphs, Frequency Tables, and Histograms Practice Source: CK-12  Search ARES for similar readings	Each group writes a 5-sentence recommendation for Mr Otieno using the claim-evidence-reasoning pattern. Claim: for example, 'Do not spend the money on a bicycle shed on this evidence; collect whole-school data before deciding on the bus or lunch serving.' Evidence: 14 of 40 walk (35%); 12 of 40 use matatu (30%); together 26 of 40 (65%); only 3 of 40 (7.5%) cycle; 4 of 40 (10%) use the school bus. Reasoning: neither walking nor matatu is more than half, so 'almost everyone walks' and 'most come by matatu' are both wrong; few cycle, so little support for a shed; one class of 40 is a small sample and cannot speak for the whole school; the single 14-seater bus serves the whole school, so Zawadi's claim 'The bus is always full' cannot be confirmed or refuted by 4 learners in one class. Groups then give a 90-second presentation to 'Mr Otieno' (the teacher or a learner in role) and use a rubric to give one 'star' and one 'step' to another group.	Model one sentence frame: 'The data shows ____ (number and percentage), which means ____, so we recommend ____.' Say: 'Your page must answer Atieno, Brian and Zawadi using numbers, not opinions.' Give WAIT TIME of 15 seconds before each group presents. Cold-call 5 learners to restate another group's evidence: 'Otieno, what number did that group use for matatu? Does it support Brian?' Probe: 'Which graph would you put first for Mr Otieno, bar graph or pie chart, and why?' (Either may be defended; expect bar graph for comparing exact counts and pie chart for showing parts of the whole.) Probe: 'What would change your recommendation?' Press for honest limits: 'What does this class data NOT tell us?' Role-play Mr Otieno asking: 'Can I build the bus route from this?'	Claim-Evidence-Reasoning with peer critique and argument from evidence: forces learners to link every conclusion to a specific number, to state what the data cannot show (sample of one class), and to defend a choice of graph or summary, building the Final Explanation.	Look for: recommendations that quote exact anchor numbers; correct statement that neither claim is more than half; correct treatment of Zawadi's claim as untestable with this data; use of 'sample' or 'whole school'; no claim that the data proves a bus route is not needed. Use the rubric levels (Exceeds, Meets, Approaches, Below): Meets requires correct data, correct graph or chart, and accurate interpretation with at most minor errors. Record the level for each group.
Driving Question Board (DQB) Creation  VIDEO: Statistics intro:	Learners return to the DQB from Lesson 1. In their groups they take each question card, decide whether it is Answered, Partly answered or Still open, and write the evidence from Lessons 1 to 7	Say: 'Go to the board and move each card. Prove it with a number.' Give WAIT TIME of 10 seconds before asking for the reason behind each move. Cold-call 4 learners: 'Brian, which card did you	DQB revisiting and sorting (Answered, Partly answered, Open): makes learning visible, ties each question to evidence, and shows that science and statistics progress by turning questions into	Indicators: every moved card has a number as evidence; learners place the bus question as open or partly open for a correct reason (single 14-seater bus, one class only); at least two new questions

Mean, median, & mode Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) Source: CK-12 Search ARES for similar readings	on a new card. Example questions: 'Who is right, Atieno or Brian?' (answered: neither; walk 35%, mataatu 30%). 'Why does the list look meaningless?' (answered: unorganised raw data; a frequency table organises it). 'Which average should we trust?' (answered: depends on data type; siblings agree at about 2, runner times differ because of one slow time). 'Is the bus always full?' (still open: needs whole-school data on the 14-seater bus). 'Should we build a bicycle shed?' (answered: not on this evidence, 7.5% cycle). New questions are added, such as 'How big a sample is fair?' and 'How can we collect the whole-school data fairly?' Each learner writes a final answer to the driving question on a card.	move to Answered? What is your evidence?' Ask: 'Which question is still open, and what data would close it?' Celebrate honest 'still open' cards: 'Good statisticians say what they do not yet know.' Photograph or copy the final DQB for the class record.	new data collection.	about sample size or whole-school collection; individual answers to the driving question mention organising, displaying and interpreting data.
Model Building Phase  VIDEO: Mean, median, & mode example Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners take out the Lesson 1 draft model of 'how data becomes evidence' and, in a different colour, revise it into a final class model. The final model shows: collect (raw list of 40) to organise (tally and frequency table, total check) to summarise (mode; mean and median for numbers, choose by data type and outliers) to display (bar graph for comparing categories, line graph for change over time, pie chart for parts of a whole, honest axes starting at 0) to interpret (claims checked against numbers) to decision (no shed on this evidence; collect whole-school data before bus or lunch). They add a feedback loop: more data leads to a better decision. Each learner then writes the Final Explanation (6-8 sentences) answering the driving question and explicitly lists three changes between their Lesson 1 model and	Say: 'Hold up your Lesson 1 model. What was missing?' Give WAIT TIME of 15 seconds, then cold-call 4 learners to name one change. Prompt: 'Where in your model does the 130 s runner time show up? Where does a truncated axis show up?' Probe: 'How can the same list look meaningless in one form and decisive in another?' Expect answers about tables and charts organising and revealing patterns. Walk around and check that the Final Explanation uses the numbers 14, 12, 26 (65%), 3 (7.5%), 4 (10%) and the mean 109.3 s versus median 106 s, in learners' own words. Collect the Final Explanation and models for assessment, and mark against the rubric.	Model comparison (Lesson 1 draft versus final model) and written Final Explanation: learners visibly revise a model with new evidence, which consolidates the full cycle and lets them articulate why the same data seems meaningless raw but decisive when organised and displayed.	Look for a final model with all five cycle steps in order plus a decision and feedback loop; at least three accurate changes from the Lesson 1 draft; a Final Explanation that (1) says neither claim is more than half, (2) explains why tables and graphs make the data decisive, (3) explains that the best summary or graph depends on the data type and question, (4) recommends no shed on this evidence and whole-school data before bus or lunch, and (5) notes the limit of one class sample. Rubric: Exceeds = all five and creative justification; Meets = all five with minor errors; Approaches = three or four; Below = two or fewer.

	the final model. They complete the summary table (I observed, I learned, I can explain).			
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D. TEACHER REFLECTION
After the lesson, check: Did each group quote exact anchor numbers (14, 12, 26 = 65%, 3 = 7.5%, 4 = 10%) rather than opinions? Did learners treat Zawadi's bus claim as untestable with one class, rather than saying it was disproved? Which groups confused mean with mode, or tried to find a mean of travel categories? Did any pie angles fail to sum to 360, or any bar graph axis start above 0? Which statistics step needs a revisit before the next unit (for example, choosing between mean and median when there is an outlier)? How well did quieter learners contribute through assigned group roles? Plan a follow-up where Grade 10 Green helps design a fair whole-school survey for Mr Otieno.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did you observe? Write what the raw list of 40 responses looked like, and what the frequency table, bar graph and pie chart for Grade 10 Green showed (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4), including the mean, median and mode of the runner times.
What did I learn?	What did you learn? Explain how organising, summarising and displaying data turns a messy list into evidence, why neither 'almost everyone walks' nor 'most come by matatu' is right, and why the best summary or graph depends on the data and the question.
How does this explain the phenomenon?	How can you explain the answer to the driving question? Write your Final Explanation for Mr Otieno: what the evidence says about the bicycle shed, school bus and lunch serving, and why he should collect whole-school data before deciding.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.